Set 2D: Op-Amps with Finite Gain-Bandwidth

ECEN 325 — Lecture 2 (Operational Amplifiers) practice set

What this set trains. Real op-amps have a finite open-loop gain $A(s) = \frac{A_0}{1+s/\omega_b}$. Instead of the virtual short you write $V_{out} = A(V_+ - V_-)$, express V_+ and V_- with dividers (superposition), and solve. For resistive (or purely capacitive) feedback the result is always the ideal gain times a single pole near $\beta\omega_t$: the gain-bandwidth trade-off. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-2 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1 tools for impedances, dividers, Bode plots and sinusoids):

T1 *Ideal op-amp with negative feedback:* no current into + or −; $V_- = V_+$ (virtual short). Its output is an ideal voltage source (loads on it do not matter).

T2 *Block formulas:* inverting $\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_{in}}$; non-inverting $\frac{V_o}{V_+} = 1 + \frac{Z_f}{Z_g}$ (with V_+ from the divider at the + input); a follower has gain 1. Several blocks in cascade: multiply.

T3 *KCL with impedances:* at a node, $\sum_k \frac{V_{node}-V_k}{Z_k} = 0$ (for a capacitor $\frac{1}{Z} = sC$). Write it at every node except op-amp outputs and source nodes.

T4 *Solving KCL equations:* pick the equation with the fewest unknowns, solve it for one node voltage, substitute into the rest; repeat until only V_o and V_i remain.

T5 *Finite gain:* $V_{out} = A(s)(V_+ - V_-)$, $A(s) = \frac{A_0}{1+s/\omega_b}$, $\omega_t = A_0\omega_b$. With $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$: $\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1+A\beta}$.

T6 *Slew rate:* for $V_{out} = \hat{V}\sin\omega t$, the fastest change is $\left|\frac{dV_{out}}{dt}\right|_{max} = \omega\hat{V}$; no distortion requires $\omega\hat{V} \le SR$.

Steps:

1. *Sources.* DC and AC both present ⇒ superposition.
2. *DC circuit:* $V_i \to 0$, $C \to$ open; pass the DC through each block with its gain at $s = 0$.
3. *Op-amp rules:* confirm negative feedback; apply T1 (or T5 for a non-ideal op-amp).
4. *Transfer function:* standard blocks → T2; anything else → T3 + T4.
5. *Standard form*, 6. *Bode*, 7. *Output* — exactly as in Lecture 1.

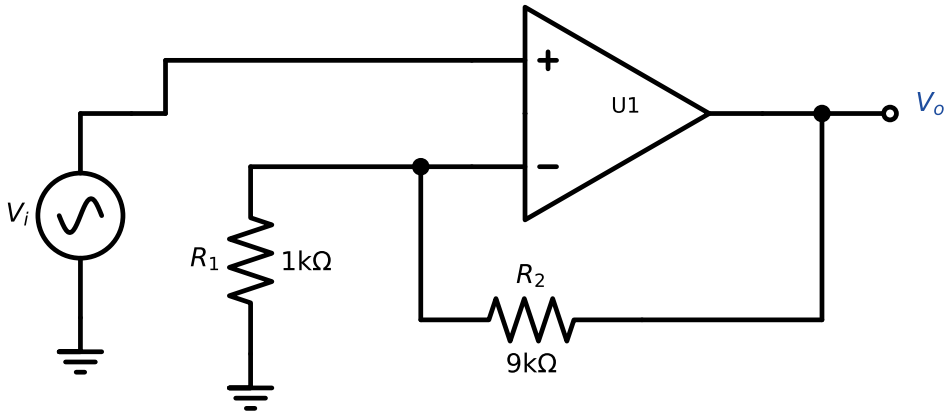
Variations: a design problem runs these steps backwards (Bode plot → $H(s)$ → blocks → component values); a slew-rate problem stops after Step 4 for every op-amp output and applies T6.

Problems

Easy problems

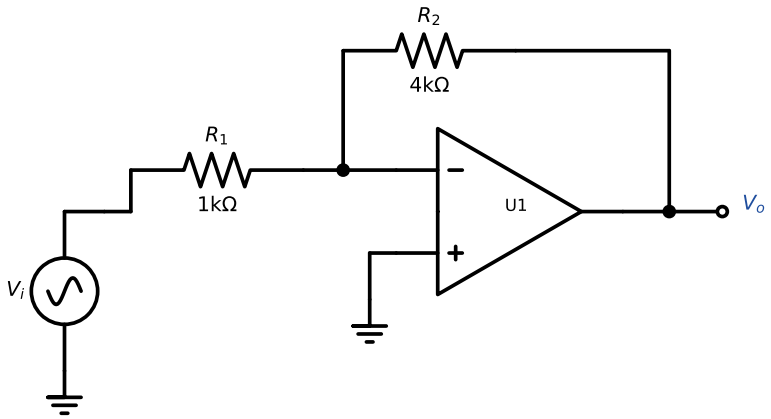
Problem 2D.1 — Easy

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin(2\pi \cdot 50000\, t)$.



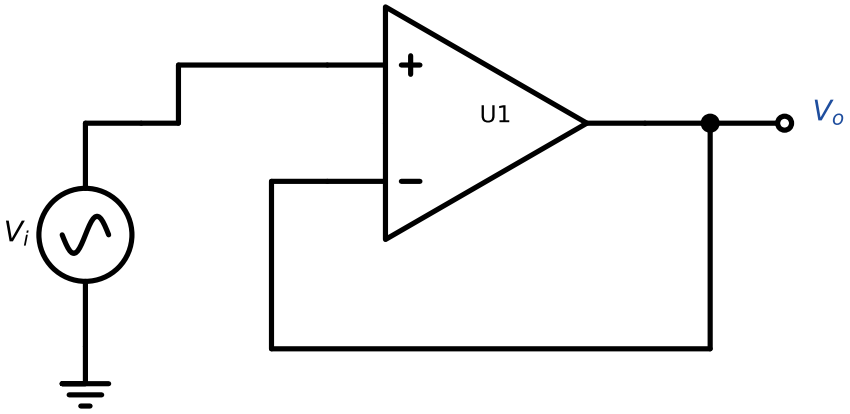
Problem 2D.2 — Easy

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{2 \times 10^5}{1 + \dfrac{s}{2\pi \cdot 5}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2 \sin\left(2\pi \cdot 10^5\, t\right)$.



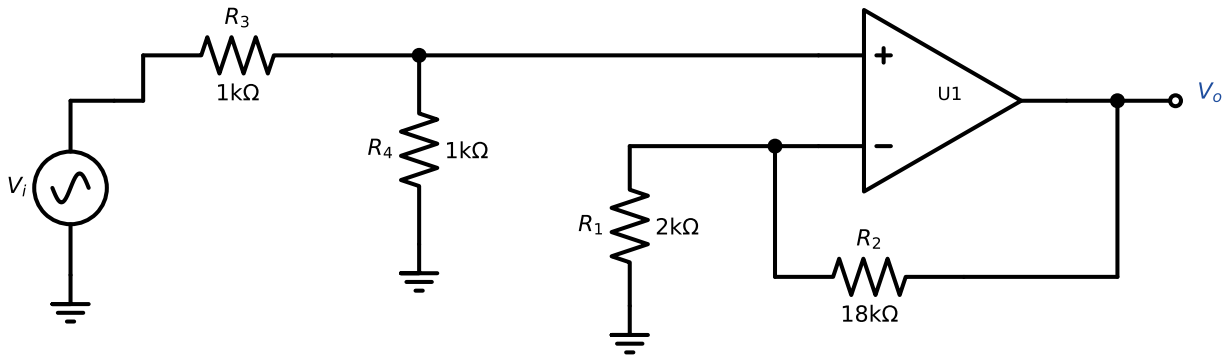
Problem 2D.3 — Easy

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 20}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1 \sin\left(2\pi \cdot 5 \times 10^5\, t\right)$.



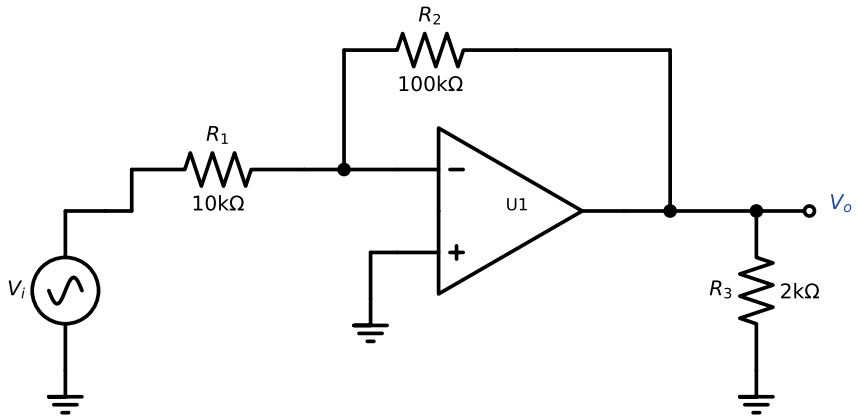
Problem 2D.4 — Easy

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2 \sin\left(2\pi \cdot 20000\, t\right)$.



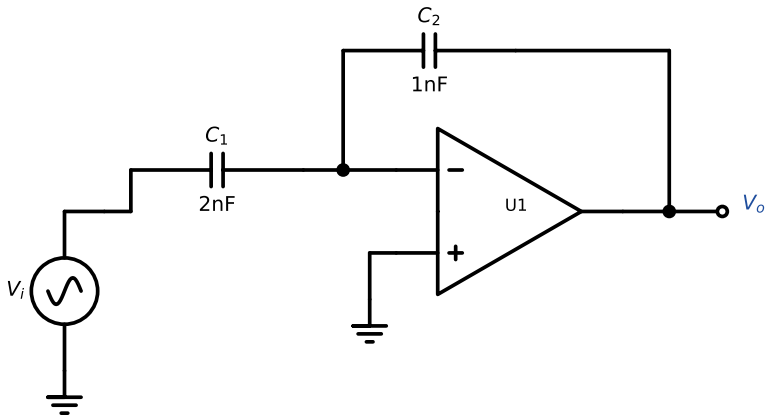
Problem 2D.5 — Easy

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \sin(2\pi \cdot 50000\, t)$.



Problem 2D.6 — Easy

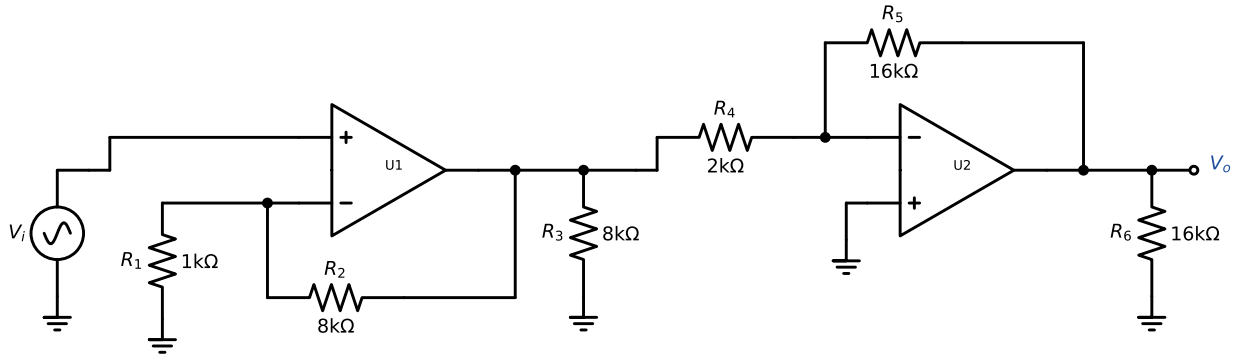
For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2 \sin\left(2\pi \cdot 2 \times 10^5\, t\right)$.



Medium problems

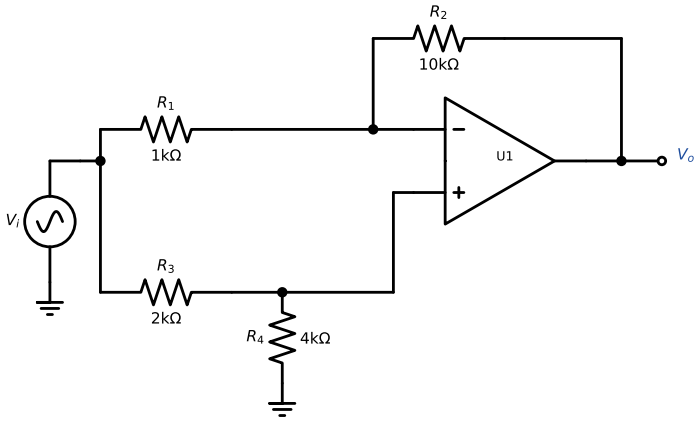
Problem 2D.7 — Medium

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 20}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.01 \sin\left(2\pi \cdot 10^5\, t\right)$.



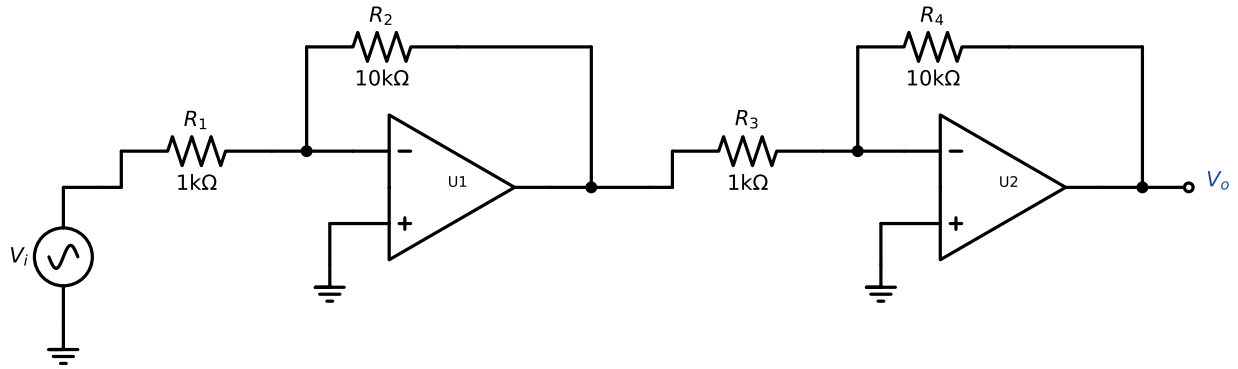
Problem 2D.8 — Medium

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin\left(2\pi \cdot 20000\, t\right)$.



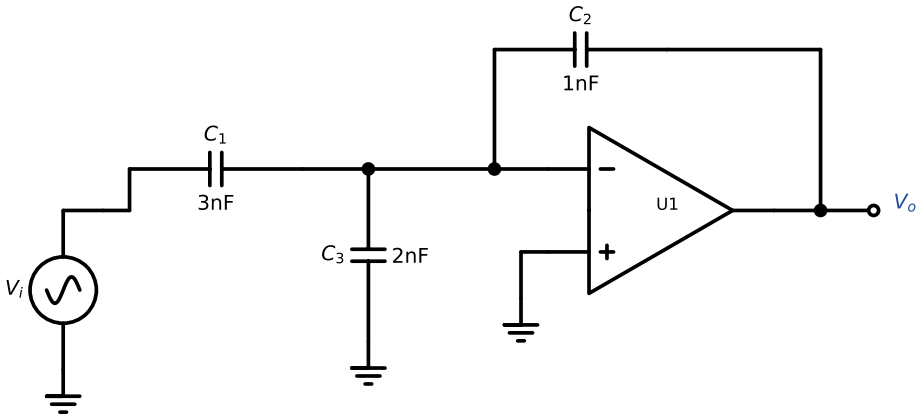
Problem 2D.9 — Medium

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.01 \sin(2\pi \cdot 50000\, t)$.



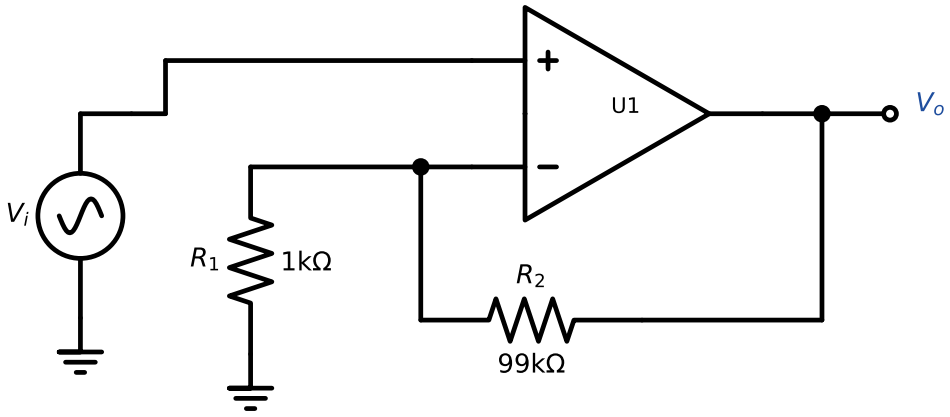
Problem 2D.10 — Medium

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 30}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \sin\left(2\pi \cdot 5 \times 10^5\, t\right)$.



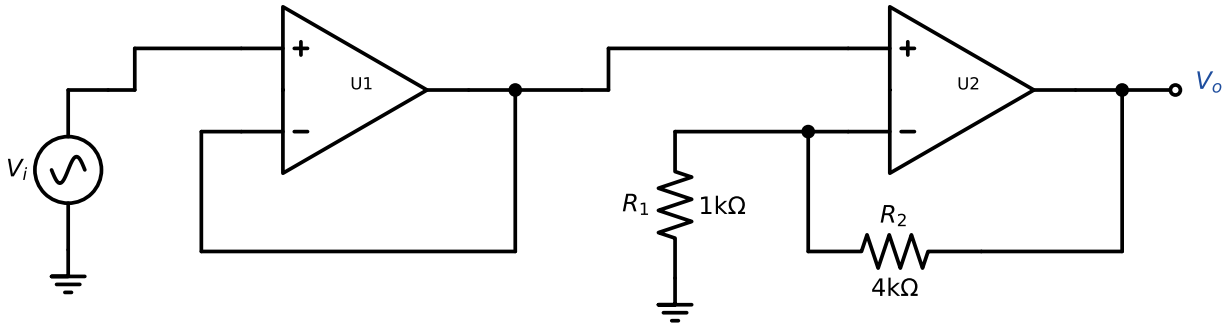
Problem 2D.11 **Medium**

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.01 \sin(2\pi \cdot 5000\, t)$.



Problem 2D.12 **Medium**

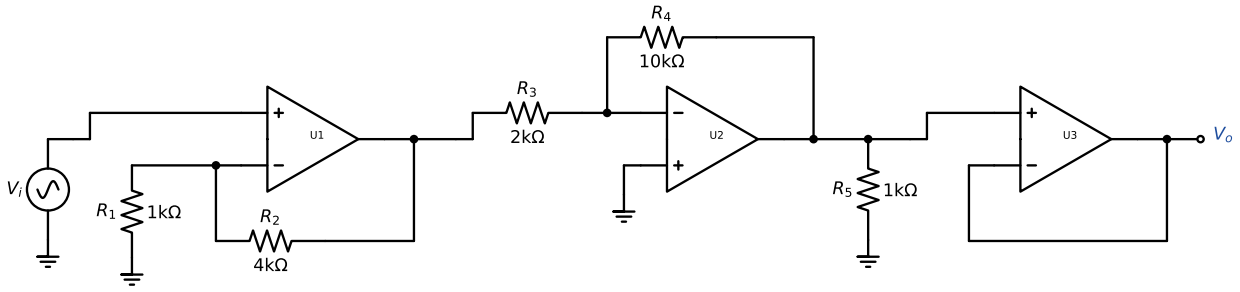
For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{2 \times 10^5}{1 + \dfrac{s}{2\pi \cdot 5}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin\left(2\pi \cdot 2 \times 10^5\, t\right)$.



Hard problems

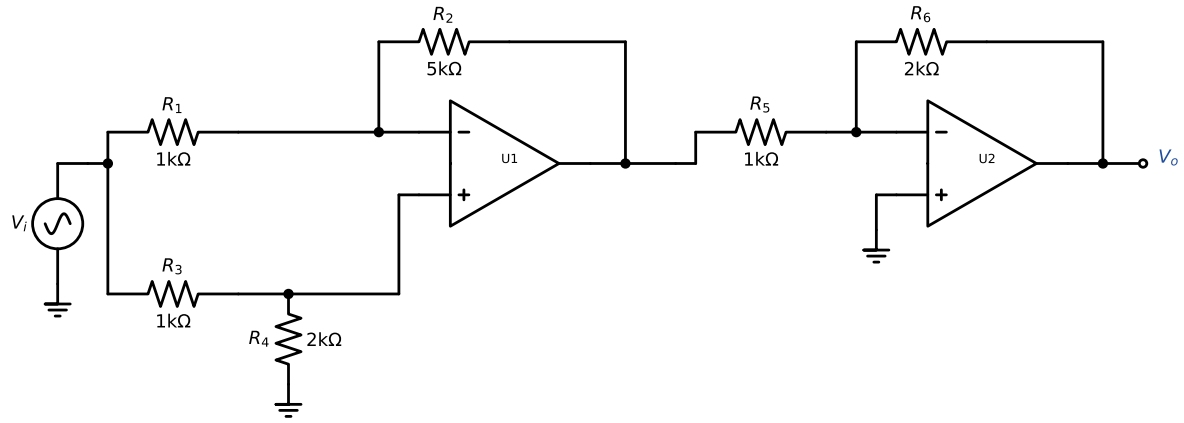
Problem 2D.13 — Hard

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.02 \sin\left(2\pi \cdot 10^5\, t\right)$.



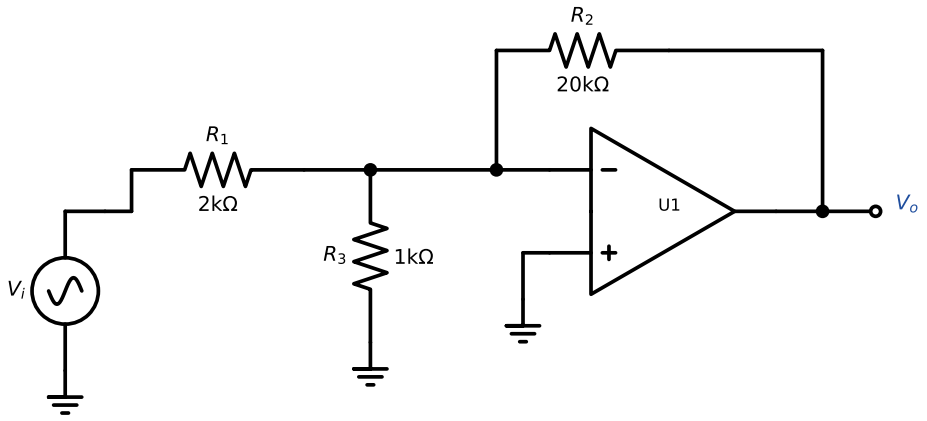
Problem 2D.14 — Hard

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 20}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \sin\left(2\pi \cdot 50000\, t + \frac{\pi}{4}\right)$.



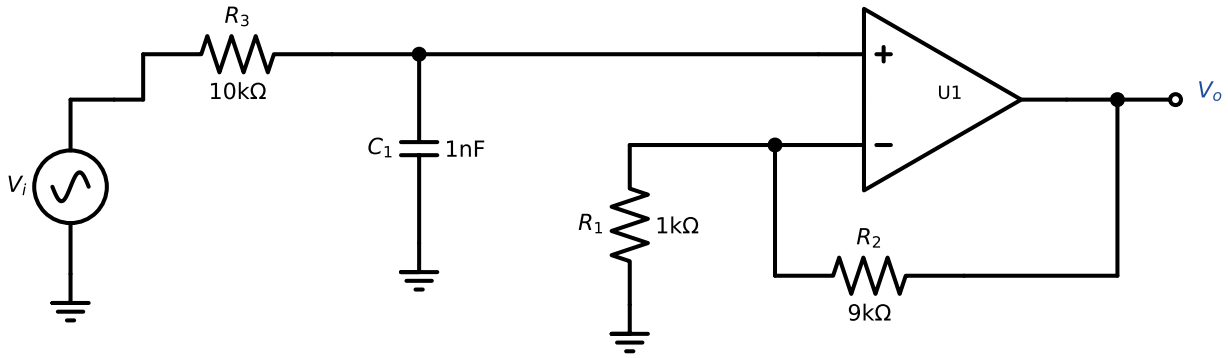
Problem 2D.15 — Hard

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \cos(2\pi \cdot 20000\, t)$.



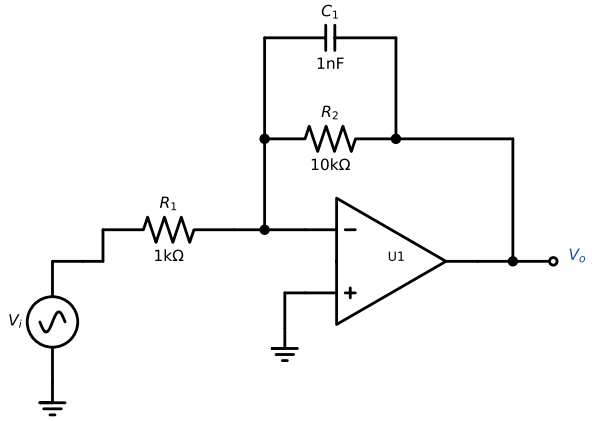
Problem 2D.16 — Hard

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin(2\pi \cdot 10000\, t)$.



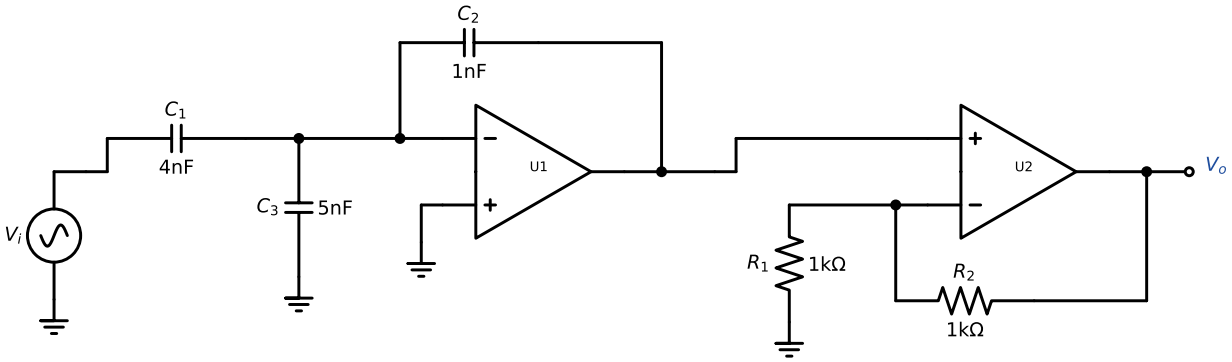
Problem 2D.17 — Hard

For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin\left(2\pi \cdot 30000\, t - \frac{\pi}{6}\right)$.



Problem 2D.18 — Hard

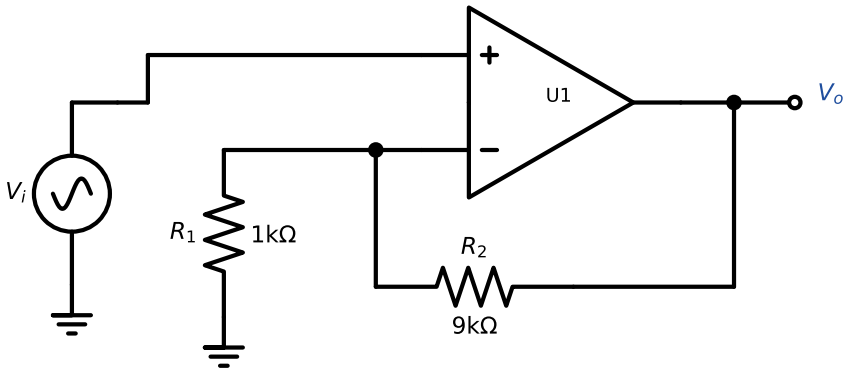
For the circuit below, where every op-amp has the open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$: (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.02 \sin\left(2\pi \cdot 10^5\, t\right)$.



Solutions

Every solution follows the steps of the Lecture-2 pipeline card.

Solution 2D.1 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = \frac{R_1}{R_1 + R_2} V_o$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 0.1$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 10$, $A_0\beta = 10000$.

$$\frac{V_o}{V_i} = 10 \cdot \frac{10000}{1 + 10000} \cdot \frac{1}{1 + \dfrac{s}{6.28 \times 10^5}} = \frac{9.999}{1 + \dfrac{s}{6.28 \times 10^5}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 6.28 \times 10^5$ rad/s, $f = 100kHz$; compare $\beta\omega_t = 6.28 \times 10^5$ rad/s.) So:

$$H(s) = \frac{6.28 \times 10^6}{s + 6.28 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 6.28 \times 10^5$; $s + 6.28 \times 10^5 = 0 \Rightarrow s = -6.28 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 6.28 \times 10^6 \, \frac{1}{(s + 6.28 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 10 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 100\text{k}}}, \qquad K = 10$$

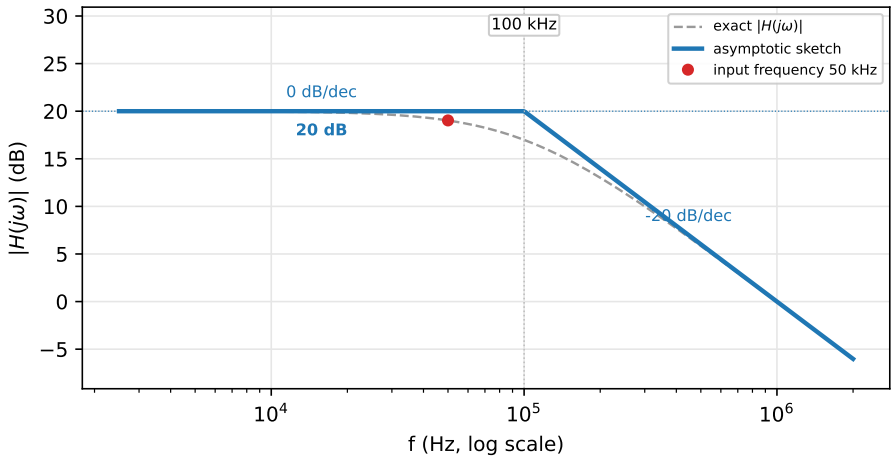
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	6.28×10^5	100 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 100$ kHz	0 dB/dec (flat)	20 dB at end
$f > 100$ kHz	-20 dB/dec	20 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 10 = 20\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 50000\,t)$, so $\omega_{in} = 2\pi{\cdot}50000 = 3.14{\times}10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	9.999	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$	0.8944	-0.464
total	**8.944** (product)	$\mathbf{-0.464}$ (sum)

$$|H(j\omega_{in})| = 8.94, \qquad \angle H(j\omega_{in}) = -0.464 \text{ rad}$$

$$v_{o,ac}(t) = 8.94 \times 0.1 \sin(2\pi \cdot 50000\,t - 0.464) = 0.894 \sin(2\pi \cdot 50000\,t - 0.464)$$

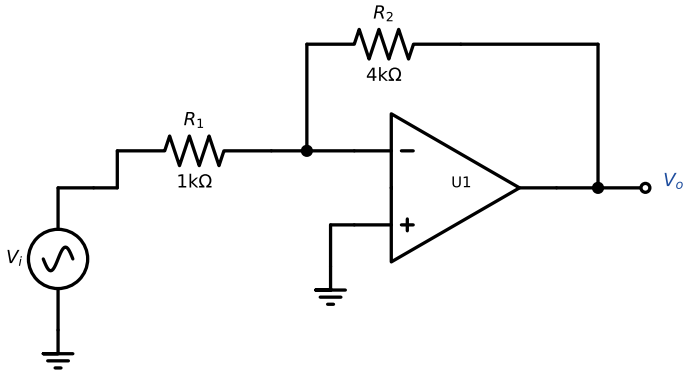
$$\boxed{V_o(t) = 0.894 \sin(2\pi \cdot 50000\,t - 0.464) \text{ V}}$$

Answer

$$H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}} \quad \text{(corners: 100 kHz)}$$

$$V_o(t) = 0.894 \sin(2\pi \cdot 50000\,t - 0.464) \text{ V}$$

Solution 2D.2 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 2 \times 10^5$, $\omega_b = 2\pi \cdot 5 = 31.4\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (superposition: V_i alone with $V_o = 0$ gives a divider R_1–R_2, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_2}{R_1 + R_2} V_i + \frac{R_1}{R_1 + R_2} V_o$$

So $\alpha_+ = 0$, $\alpha_- = 0.8$, $\beta = 0.2$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -4$, $A_0\beta = 40000$.

$$\frac{V_o}{V_i} = -4 \cdot \frac{40000}{1 + 40000} \cdot \frac{1}{1 + \dfrac{s}{1.26 \times 10^6}} = \frac{-4}{1 + \dfrac{s}{1.26 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 1.26 \times 10^6$ rad/s, $f = 200kHz$; compare $\beta\omega_t = 1.26 \times 10^6$ rad/s.) So:

$$H(s) = \frac{-5.03 \times 10^6}{s + 1.26 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 1.26 \times 10^6$; $s + 1.26 \times 10^6 = 0 \Rightarrow s = -1.26 \times 10^6$.

Factored form (leading coefficient in front):

$$H(s) = -5.03 \times 10^6 \, \frac{1}{(s + 1.26 \times 10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -4 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 200\text{k}}}, \qquad K = -4$$

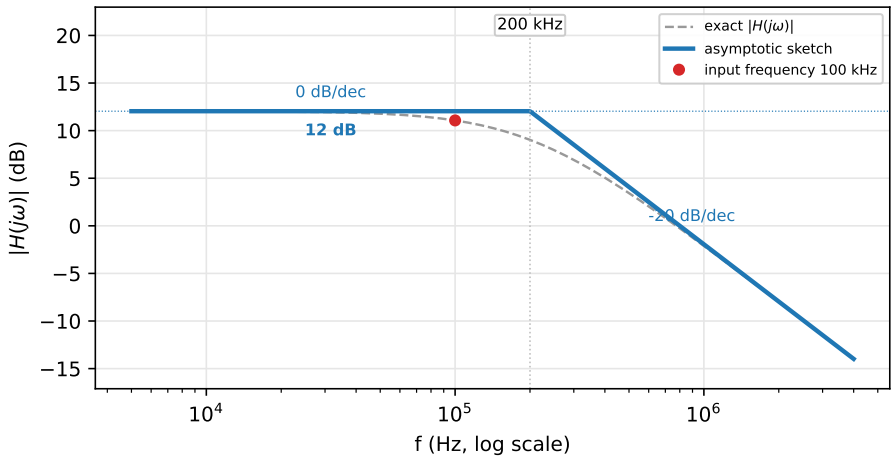
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.26×10^6	200 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 4 = 12\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 200$ kHz	0 dB/dec (flat)	12 dB at end
$f > 200$ kHz	-20 dB/dec	12 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 4 = 12\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin\left(2\pi \cdot 10^5\, t\right)$, so $\omega_{in} = 2\pi \cdot 10^5 = 6.28 \times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-4	4	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 200\text{k}}}$	0.8944	-0.464
total	**3.578** (product)	$-$**3.61** (sum)

$$|H(j\omega_{in})| = 3.58, \qquad \angle H(j\omega_{in}) = -3.61 \text{ rad}$$

$$v_{o,ac}(t) = 3.58 \times 0.2 \sin\left(2\pi \cdot 10^5\, t - 3.61\right) = 0.716 \sin\left(2\pi \cdot 10^5\, t - 3.61\right)$$

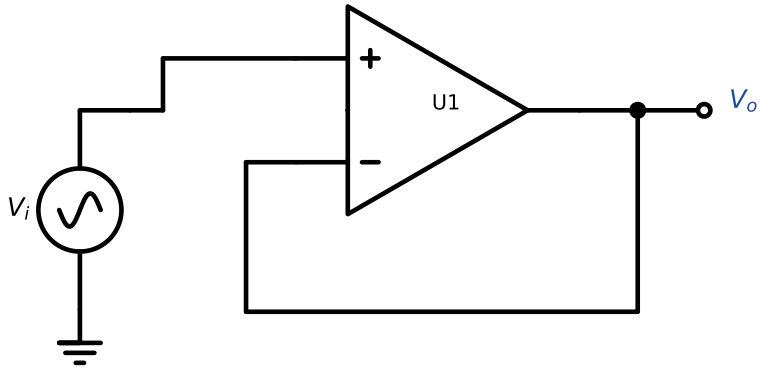
$$\boxed{V_o(t) = 0.716 \sin\left(2\pi \cdot 10^5\, t - 3.61\right) \text{ V}}$$

Answer

$$H(s) = -4\,\frac{1}{1+\dfrac{s}{2\pi \cdot 200\text{k}}} \quad \text{(corners: 200 kHz)}$$

$$V_o(t) = 0.716 \sin\left(2\pi \cdot 10^5\, t - 3.61\right) \text{ V}$$

Solution 2D.3 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 20 = 126\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 1.26 \times 10^7\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = V_o$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 1$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 1$, $A_0\beta = 10^5$.

$$\frac{V_o}{V_i} = 1 \cdot \frac{10^5}{1 + 10^5} \cdot \frac{1}{1 + \dfrac{s}{1.26 \times 10^7}} = \frac{1}{1 + \dfrac{s}{1.26 \times 10^7}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 1.26 \times 10^7\,\text{rad/s}$, $f = 2MHz$; compare $\beta\omega_t = 1.26 \times 10^7\,\text{rad/s}$.) So:

$$H(s) = \frac{1.26 \times 10^7}{s + 1.26 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 1.26 \times 10^7$; $s + 1.26 \times 10^7 = 0 \Rightarrow s = -1.26 \times 10^7$.

Factored form (leading coefficient in front):

$$H(s) = 1.26 \times 10^7 \, \frac{1}{(s + 1.26 \times 10^7)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 2\mathrm{M}}}, \qquad K = 1$$

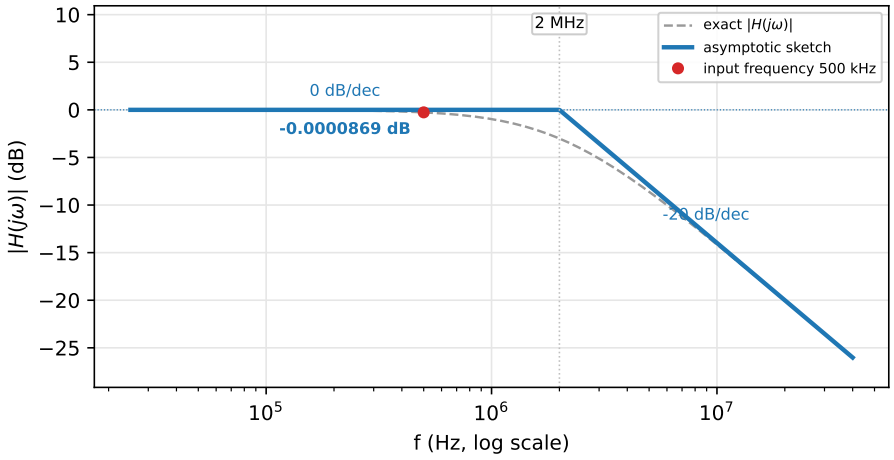
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.26×10^7	2 MHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = -0.0000869\,\mathrm{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 2$ MHz	0 dB/dec (flat)	-0.0000869 dB at end
$f > 2$ MHz	-20 dB/dec	-0.0000869 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1 = -0.0000869\,\mathrm{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin\left(2\pi\cdot 5\times 10^5\, t\right)$, so $\omega_{in} = 2\pi\cdot 5\times 10^5 = 3.14\times 10^6$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{M}}}$	0.9701	-0.245
total	**0.9701** (product)	$-$**0.245** (sum)

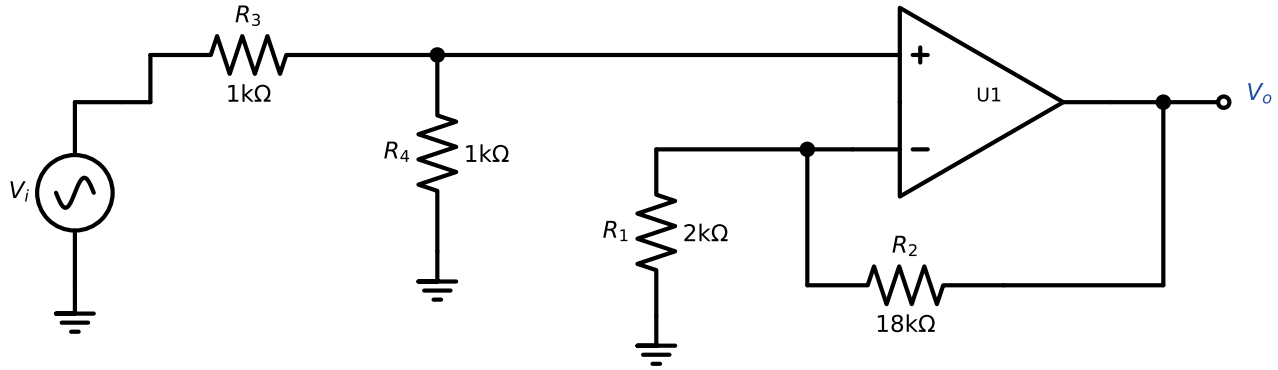
$$|H(j\omega_{in})| = 0.97, \qquad \angle H(j\omega_{in}) = -0.245 \text{ rad}$$
$$v_{o,ac}(t) = 0.97\times 1\sin\left(2\pi\cdot 5\times 10^5\, t - 0.245\right) = 0.97\sin\left(2\pi\cdot 5\times 10^5\, t - 0.245\right)$$

$$\boxed{V_o(t) = 0.97\sin\left(2\pi\cdot 5\times 10^5\, t - 0.245\right)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{M}}} \quad \text{(corners: 2 MHz)}$$
$$V_o(t) = 0.97\sin\left(2\pi\cdot 5\times 10^5\, t - 0.245\right)\ \text{V}$$

Solution 2D.4 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = \frac{R_4}{R_3 + R_4} V_i$$
$$V_- = \frac{R_1}{R_1 + R_2} V_o$$

So $\alpha_+ = 0.5$, $\alpha_- = 0$, $\beta = 0.1$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 5$, $A_0\beta = 10000$.

$$\frac{V_o}{V_i} = 5 \cdot \frac{10000}{1 + 10000} \cdot \frac{1}{1 + \dfrac{s}{6.28 \times 10^5}} = \frac{5}{1 + \dfrac{s}{6.28 \times 10^5}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 6.28 \times 10^5$ rad/s, $f = 100kHz$; compare $\beta\omega_t = 6.28 \times 10^5$ rad/s.) So:

$$H(s) = \frac{3.14 \times 10^6}{s + 6.28 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 6.28 \times 10^5$; $s + 6.28 \times 10^5 = 0 \Rightarrow s = -6.28 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 3.14 \times 10^6 \, \frac{1}{(s + 6.28 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 5 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 100\text{k}}}, \qquad K = 5$$

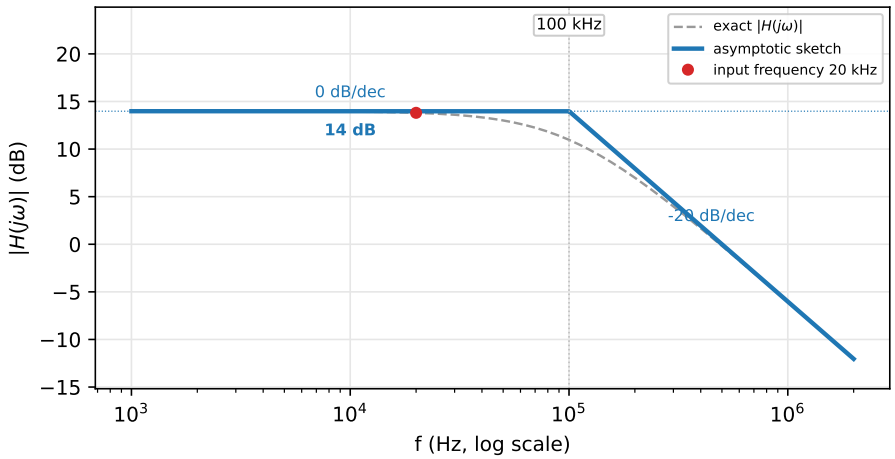
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	6.28×10^5	100 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 5 = 14\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 100$ kHz	0 dB/dec (flat)	14 dB at end
$f > 100$ kHz	-20 dB/dec	14 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 20000\,t)$, so $\omega_{in} = 2\pi{\cdot}20000 = 1.26{\times}10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5	5	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$	0.9806	-0.197
total	**4.902** (product)	$\mathbf{-0.197}$ (sum)

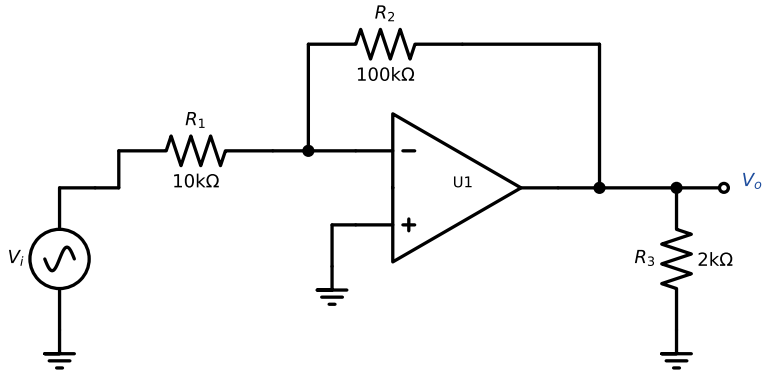
$$|H(j\omega_{in})| = 4.9, \qquad \angle H(j\omega_{in}) = -0.197 \text{ rad}$$
$$v_{o,ac}(t) = 4.9 \times 0.2 \sin(2\pi \cdot 20000\,t - 0.197) = 0.98 \sin(2\pi \cdot 20000\,t - 0.197)$$

$$\boxed{V_o(t) = 0.98 \sin(2\pi \cdot 20000\,t - 0.197)\ \text{V}}$$

Answer

$$H(s) = 5\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}} \quad \text{(corners: 100 kHz)}$$
$$V_o(t) = 0.98 \sin(2\pi \cdot 20000\,t - 0.197)\ \text{V}$$

Solution 2D.5 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

The load R_3 sits on an op-amp output, modelled as the ideal source $A(V_+ - V_-)$: it does not change the output voltage and drops out. **Block 1 (U1).** (superposition: V_i alone with $V_o = 0$ gives a divider R_1–R_2, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_2}{R_1 + R_2} V_i + \frac{R_1}{R_1 + R_2} V_o$$

So $\alpha_+ = 0$, $\alpha_- = 0.9091$, $\beta = 0.09091$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -10$, $A_0\beta = 9090$.

$$\frac{V_o}{V_i} = -10 \cdot \frac{9090}{1 + 9090} \cdot \frac{1}{1 + \dfrac{s}{5.71 \times 10^5}} = \frac{-9.999}{1 + \dfrac{s}{5.71 \times 10^5}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 5.71 \times 10^5$ rad/s, $f = 90.9kHz$; compare $\beta\omega_t = 5.71 \times 10^5$ rad/s.) So:

$$H(s) = \frac{-5.71 \times 10^6}{s + 5.71 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 5.71 \times 10^5$; $s + 5.71 \times 10^5 = 0 \Rightarrow s = -5.71 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = -5.71 \times 10^6 \, \frac{1}{(s + 5.71 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -10 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 90.9\text{k}}}, \qquad K = -10$$

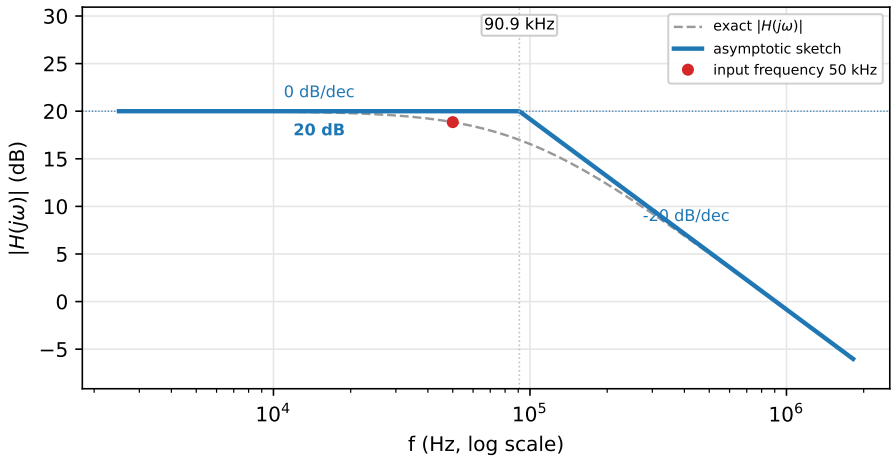
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	5.71×10^5	90.9 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 90.9$ kHz	0 dB/dec (flat)	20 dB at end
$f > 90.9$ kHz	-20 dB/dec	20 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 10 = 20\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi \cdot 50000\,t)$, so $\omega_{in} = 2\pi \cdot 50000 = 3.14 \times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-10	9.999	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 90.9\text{k}}}$	0.8762	-0.503
total	**8.761** (product)	$-$**3.64** (sum)

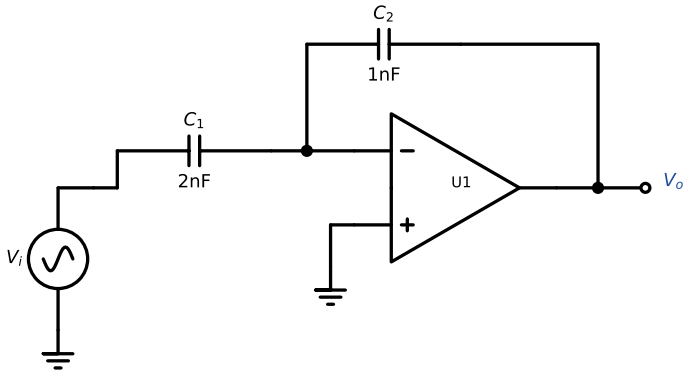
$$|H(j\omega_{in})| = 8.76, \qquad \angle H(j\omega_{in}) = -3.64 \text{ rad}$$
$$v_{o,ac}(t) = 8.76 \times 0.05 \sin(2\pi \cdot 50000\,t - 3.64) = 0.438 \sin(2\pi \cdot 50000\,t - 3.64)$$

$$\boxed{V_o(t) = 0.438 \sin(2\pi \cdot 50000\,t - 3.64) \text{ V}}$$

Answer

$$H(s) = -10\,\frac{1}{1+\dfrac{s}{2\pi \cdot 90.9\text{k}}} \quad \text{(corners: 90.9 kHz)}$$
$$V_o(t) = 0.438 \sin(2\pi \cdot 50000\,t - 3.64) \text{ V}$$

Solution 2D.6 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (superposition: V_i alone with $V_o = 0$ gives a divider $\frac{1}{sC_1} - \frac{1}{sC_2}$, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{\frac{1}{sC_2}}{\frac{1}{sC_1} + \frac{1}{sC_2}} V_i + \frac{\frac{1}{sC_1}}{\frac{1}{sC_1} + \frac{1}{sC_2}} V_o$$

(Only capacitors: every impedance is $\frac{1}{sC}$, so in each ratio the s cancels, e.g. $\frac{1/(sC_a)}{1/(sC_a)+1/(sC_b)} = \frac{C_b}{C_a+C_b}$, and $\frac{1}{sC_a} \parallel \frac{1}{sC_b} = \frac{1}{s(C_a+C_b)}$.) So $\alpha_+ = 0$, $\alpha_- = 0.6667$, $\beta = 0.3333$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -2$, $A_0\beta = 33300$.

$$\frac{V_o}{V_i} = -2 \cdot \frac{33300}{1 + 33300} \cdot \frac{1}{1 + \dfrac{s}{2.09 \times 10^6}} = \frac{-2}{1 + \dfrac{s}{2.09 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 2.09 \times 10^6$ rad/s, $f = 333kHz$; compare $\beta\omega_t = 2.09 \times 10^6$ rad/s.) So:

$$H(s) = \frac{-4.19 \times 10^6}{s + 2.09 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 2.09 \times 10^6$; $s + 2.09 \times 10^6 = 0 \Rightarrow s = -2.09 \times 10^6$.

Factored form (leading coefficient in front):

$$H(s) = -4.19 \times 10^6 \, \frac{1}{(s + 2.09 \times 10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -2 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 333\text{k}}}, \qquad K = -2$$

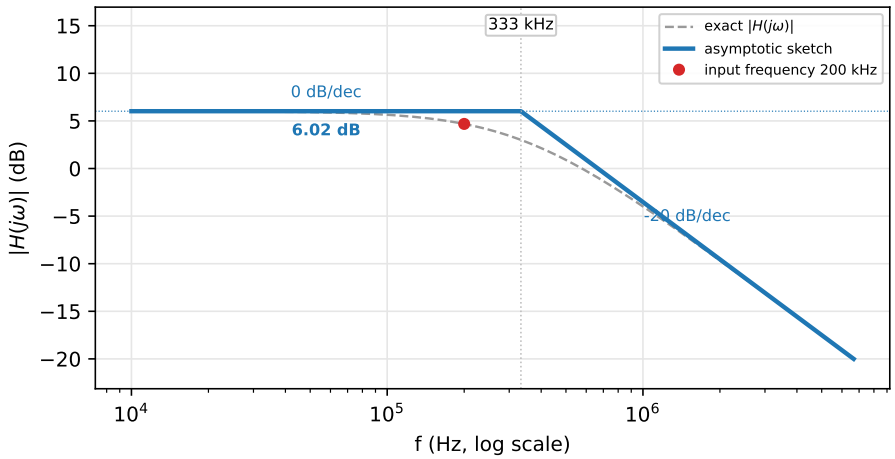
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2.09×10^6	333 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 2 = 6.02$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 333$ kHz	0 dB/dec (flat)	6.02 dB at end
$f > 333$ kHz	-20 dB/dec	6.02 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 2 = 6.02$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin\left(2\pi \cdot 2\times 10^5\, t\right)$, so $\omega_{in} = 2\pi \cdot 2\times 10^5 = 1.26\times 10^6$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-2	2	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 333\text{k}}}$	0.8575	-0.54
total	**1.715** (product)	$\mathbf{-3.68}$ (sum)

$$|H(j\omega_{in})| = 1.71, \qquad \angle H(j\omega_{in}) = -3.68 \text{ rad}$$

$$v_{o,ac}(t) = 1.71\times 0.2\sin\left(2\pi\cdot 2\times 10^5\, t - 3.68\right) = 0.343\sin\left(2\pi\cdot 2\times 10^5\, t - 3.68\right)$$

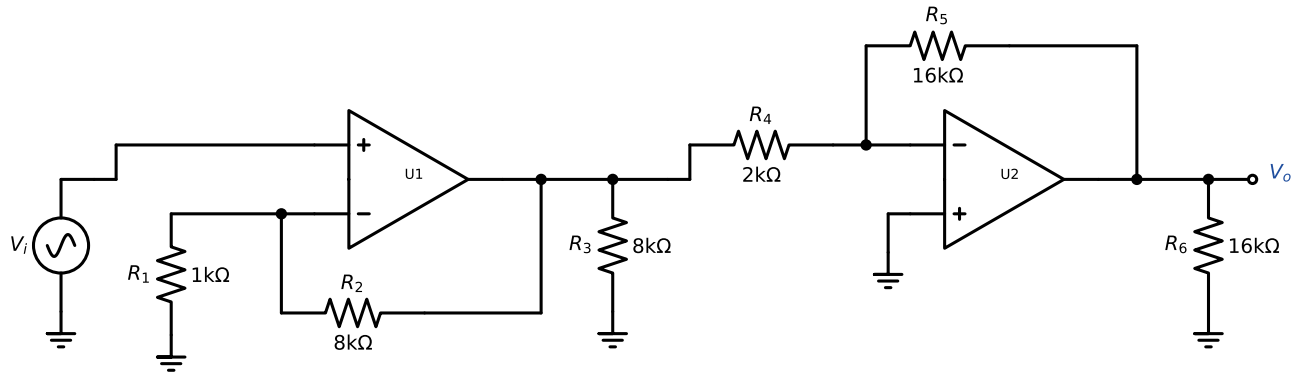
$$\boxed{V_o(t) = 0.343\sin\left(2\pi\cdot 2\times 10^5\, t - 3.68\right)\ \text{V}}$$

Answer

$$H(s) = -2\,\frac{1}{1+\dfrac{s}{2\pi\cdot 333\text{k}}} \quad \text{(corners: 333 kHz)}$$

$$V_o(t) = 0.343\sin\left(2\pi\cdot 2\times 10^5\, t - 3.68\right)\ \text{V}$$

Solution 2D.7 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 20 = 126\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 1.26 \times 10^7\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

The load R_3, R_6 sits on an op-amp output, modelled as the ideal source $A(V_+ - V_-)$: it does not change the output voltage and drops out. **Block 1 (U1).** (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = \frac{R_1}{R_1 + R_2} V_{o1}$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 0.1111$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 9$, $A_0\beta = 11100$.

$$\frac{V_{o1}}{V_i} = 9 \cdot \frac{11100}{1 + 11100} \cdot \frac{1}{1 + \dfrac{s}{1.4 \times 10^6}} = \frac{8.999}{1 + \dfrac{s}{1.4 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 1.4 \times 10^6$ rad/s, $f = 222kHz$; compare $\beta\omega_t = 1.4 \times 10^6$ rad/s.)
Block 2 (U2). (superposition: V_{o1} alone with $V_o = 0$ gives a divider R_4–R_5, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_5}{R_4 + R_5} V_{o1} + \frac{R_4}{R_4 + R_5} V_o$$

So $\alpha_+ = 0$, $\alpha_- = 0.8889$, $\beta = 0.1111$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -8$, $A_0\beta = 11100$.

$$\frac{V_o}{V_{o1}} = -8 \cdot \frac{11100}{1 + 11100} \cdot \frac{1}{1 + \dfrac{s}{1.4 \times 10^6}} = \frac{-7.999}{1 + \dfrac{s}{1.4 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 1.4 \times 10^6$ rad/s, $f = 222kHz$; compare $\beta\omega_t = 1.4 \times 10^6$ rad/s.)
Each op-amp output is still an ideal source (it is $A(V_+ - V_-)$), so the stages multiply. Multiply the blocks and simplify:

$$H(s) = \frac{-1.4 \times 10^{14}}{s^2 + 2.79 \times 10^6\, s + 1.95 \times 10^{12}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 2.79 \times 10^6\, s + 1.95 \times 10^{12}$; quadratic formula:

$$s = \frac{-2.79 \times 10^6 \pm \sqrt{(2.79 \times 10^6)^2 - 4(1.95 \times 10^{12})}}{2} = \frac{-2.79 \times 10^6 \pm 0}{2} \Rightarrow s = -1.4 \times 10^6,\ -1.4 \times 10^6$$

Factored form (leading coefficient in front):

$$H(s) = -1.4 \times 10^{14} \frac{1}{(s + 1.4 \times 10^6)(s + 1.4 \times 10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -72 \frac{1}{1 + \dfrac{s}{2\pi \cdot 222\text{k}}} \frac{1}{1 + \dfrac{s}{2\pi \cdot 222\text{k}}}, \qquad K = -72$$

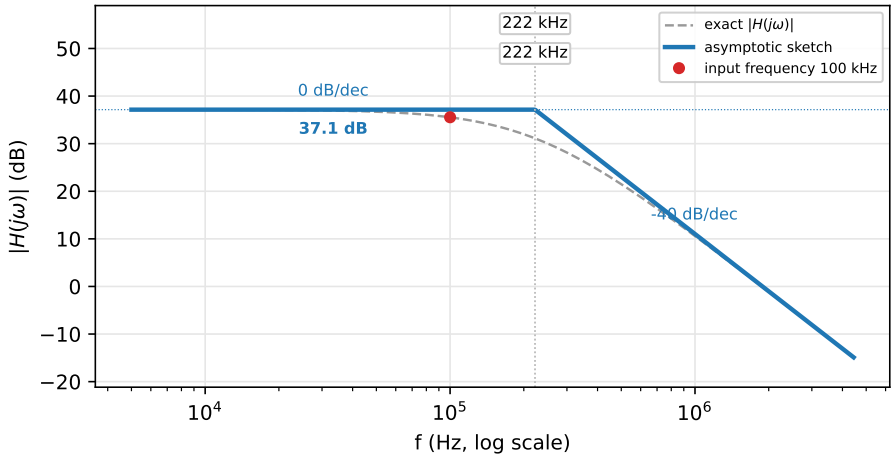
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.4×10^6	222 kHz	−20 dB/dec
pole	1.4×10^6	222 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 72 = 37.1$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 222$ kHz	0 dB/dec (flat)	37.1 dB at end
$f > 222$ kHz	-40 dB/dec	37.1 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 72 = 37.1\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin\left(2\pi\cdot 10^5\, t\right)$, so $\omega_{in} = 2\pi\cdot 10^5 = 6.28\times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-72	71.99	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 222\text{k}}}$	0.9119	-0.423
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 222\text{k}}}$	0.9119	-0.423
total	**59.87** (product)	$\mathbf{-3.99}$ (sum)

$$|H(j\omega_{in})| = 59.9, \qquad \angle H(j\omega_{in}) = -3.99 \text{ rad}$$
$$v_{o,ac}(t) = 59.9\times 0.01\,\sin\left(2\pi\cdot 10^5\, t - 3.99\right) = 0.599\,\sin\left(2\pi\cdot 10^5\, t - 3.99\right)$$

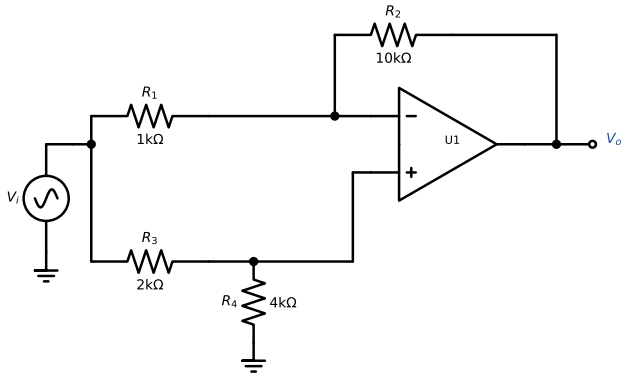
$$\boxed{V_o(t) = 0.599\,\sin\left(2\pi\cdot 10^5\, t - 3.99\right)\ \text{V}}$$

Answer

$$H(s) = -72\,\frac{1}{1+\dfrac{s}{2\pi\cdot 222\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 222\text{k}}} \quad \text{(corners: 222 kHz, 222 kHz)}$$

$$V_o(t) = 0.599\,\sin\left(2\pi\cdot 10^5\,t - 3.99\right)\ \text{V}$$

Solution 2D.8 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (each input is a divider; the − input sees both sources, use superposition)

$$V_+ = \frac{R_4}{R_3 + R_4} V_i$$
$$V_- = \frac{R_2}{R_1 + R_2} V_i + \frac{R_1}{R_1 + R_2} V_o$$

So $\alpha_+ = 0.6667$, $\alpha_- = 0.9091$, $\beta = 0.09091$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -2.667$, $A_0\beta = 9090$.

$$\frac{V_o}{V_i} = -2.667 \cdot \frac{9090}{1 + 9090} \cdot \frac{1}{1 + \dfrac{s}{5.71 \times 10^5}} = \frac{-2.666}{1 + \dfrac{s}{5.71 \times 10^5}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 5.71 \times 10^5$ rad/s, $f = 90.9kHz$; compare $\beta\omega_t = 5.71 \times 10^5$ rad/s.) So:

$$H(s) = \frac{-1.52 \times 10^6}{s + 5.71 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 5.71 \times 10^5$; $s + 5.71 \times 10^5 = 0 \Rightarrow s = -5.71 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = -1.52 \times 10^6 \, \frac{1}{(s + 5.71 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -2.67 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 90.9\text{k}}}, \qquad K = -2.67$$

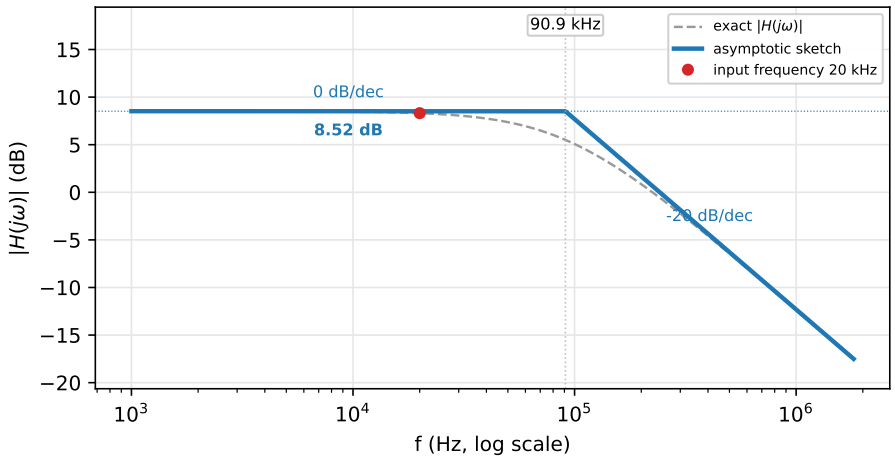
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	5.71×10^5	90.9 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 2.67 = 8.52\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 90.9$ kHz	0 dB/dec (flat)	8.52 dB at end
$f > 90.9$ kHz	−20 dB/dec	8.52 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 2.67 = 8.52\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 20000\,t)$, so $\omega_{in} = 2\pi{\cdot}20000 = 1.26{\times}10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-2.67	2.666	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}}$	0.9766	-0.217
total	**2.604** (product)	$-$**3.36** (sum)

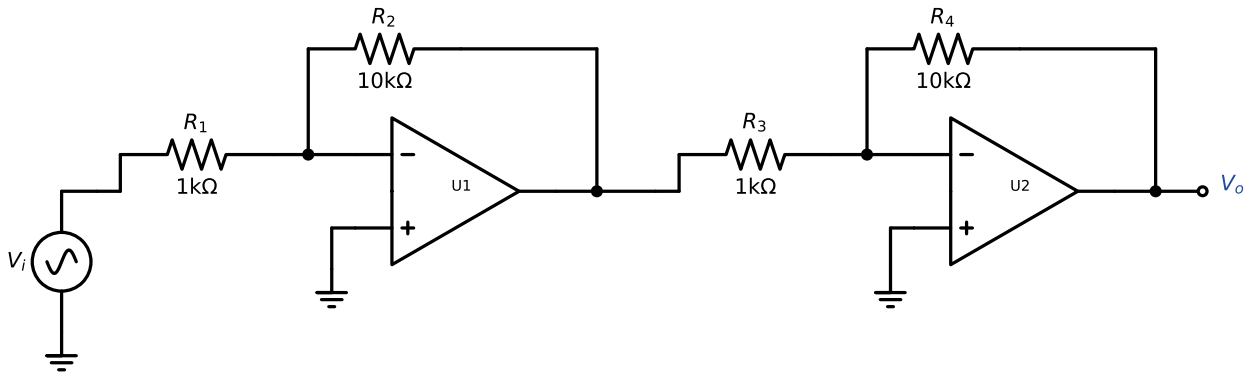
$$|H(j\omega_{in})| = 2.6, \qquad \angle H(j\omega_{in}) = -3.36 \text{ rad}$$
$$v_{o,ac}(t) = 2.6 \times 0.1 \sin(2\pi \cdot 20000\,t - 3.36) = 0.26 \sin(2\pi \cdot 20000\,t - 3.36)$$

$$\boxed{V_o(t) = 0.26 \sin(2\pi \cdot 20000\,t - 3.36) \text{ V}}$$

Answer

$$H(s) = -2.67\,\frac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}} \quad \text{(corners: 90.9 kHz)}$$
$$V_o(t) = 0.26 \sin(2\pi \cdot 20000\,t - 3.36) \text{ V}$$

Solution 2D.9 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (superposition: V_i alone with $V_{o1} = 0$ gives a divider R_1–R_2, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_2}{R_1 + R_2} V_i + \frac{R_1}{R_1 + R_2} V_{o1}$$

So $\alpha_+ = 0$, $\alpha_- = 0.9091$, $\beta = 0.09091$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -10$, $A_0\beta = 9090$.

$$\frac{V_{o1}}{V_i} = -10 \cdot \frac{9090}{1 + 9090} \cdot \frac{1}{1 + \dfrac{s}{5.71 \times 10^5}} = \frac{-9.999}{1 + \dfrac{s}{5.71 \times 10^5}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 5.71\times10^5$ rad/s, $f = 90.9kHz$; compare $\beta\omega_t = 5.71\times10^5$ rad/s.)
Block 2 (U2). (superposition: V_{o1} alone with $V_o = 0$ gives a divider R_3–R_4, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_4}{R_3+R_4}V_{o1} + \frac{R_3}{R_3+R_4}V_o$$

So $\alpha_+ = 0$, $\alpha_- = 0.9091$, $\beta = 0.09091$: ideal gain $G = \frac{\alpha_+-\alpha_-}{\beta} = -10$, $A_0\beta = 9090$.

$$\frac{V_o}{V_{o1}} = -10\cdot\frac{9090}{1+9090}\cdot\frac{1}{1+\dfrac{s}{5.71\times10^5}} = \frac{-9.999}{1+\dfrac{s}{5.71\times10^5}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 5.71\times10^5$ rad/s, $f = 90.9kHz$; compare $\beta\omega_t = 5.71\times10^5$ rad/s.)
Each op-amp output is still an ideal source (it is $A(V_+ - V_-)$), so the stages multiply. Multiply the blocks and simplify:

$$H(s) = \frac{3.26\times10^{13}}{s^2 + 1.14\times10^6\, s + 3.26\times10^{11}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 1.14\times10^6\, s + 3.26\times10^{11}$; quadratic formula:

$$s = \frac{-1.14\times10^6 \pm \sqrt{(1.14\times10^6)^2 - 4(3.26\times10^{11})}}{2} = \frac{-1.14\times10^6 \pm 0}{2} \Rightarrow s = -5.71\times10^5,\ -5.71\times10^5$$

Factored form (leading coefficient in front):

$$H(s) = 3.26\times10^{13}\,\frac{1}{(s+5.71\times10^5)(s+5.71\times10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 100\,\frac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}}, \qquad K = 100$$

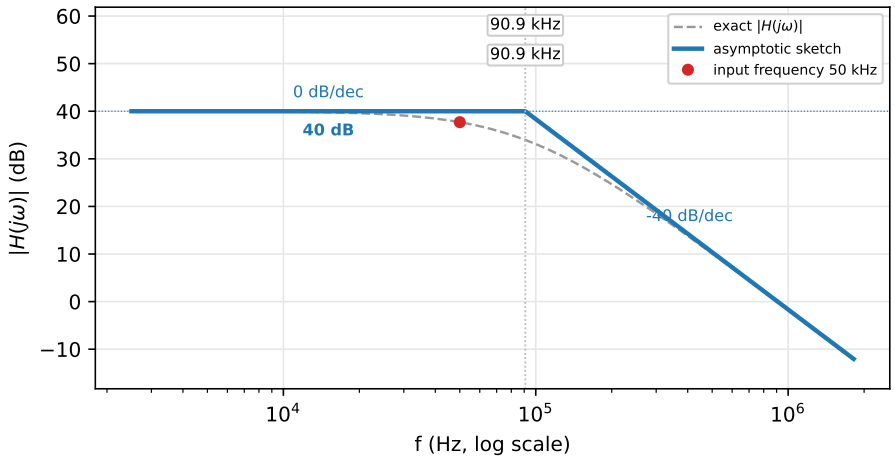
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	5.71×10^5	90.9 kHz	−20 dB/dec
pole	5.71×10^5	90.9 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 100 = 40$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 90.9$ kHz	0 dB/dec (flat)	40 dB at end
90.9 kHz – 90.9 kHz	-20 dB/dec	40 dB at start, 40 dB at end
$f > 90.9$ kHz	-40 dB/dec	40 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 100 = 40\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi\cdot 50000\,t)$, so $\omega_{in} = 2\pi\cdot 50000 = 3.14\times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
100	99.98	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}}$	0.8762	-0.503
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}}$	0.8762	-0.503
total	**76.76** (product)	$\mathbf{-1.01}$ (sum)

$$|H(j\omega_{in})| = 76.8, \qquad \angle H(j\omega_{in}) = -1.01 \text{ rad}$$
$$v_{o,ac}(t) = 76.8\times 0.01\sin(2\pi\cdot 50000\,t - 1.01) = 0.768\sin(2\pi\cdot 50000\,t - 1.01)$$

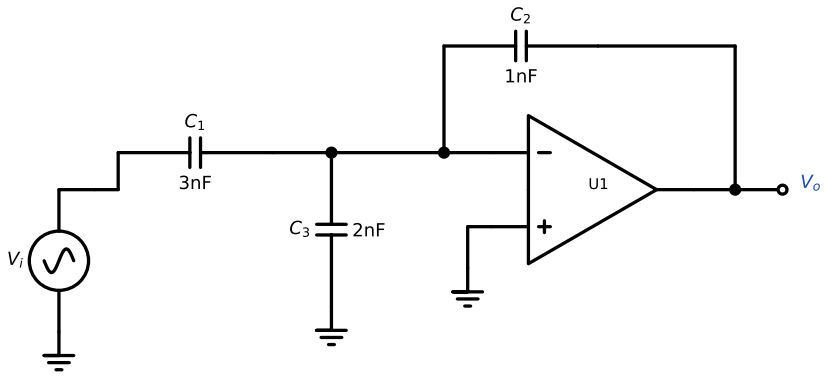
$$\boxed{V_o(t) = 0.768\sin(2\pi\cdot 50000\,t - 1.01)\ \text{V}}$$

Answer

$$H(s) = 100\,\frac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 90.9\text{k}}} \quad \text{(corners: 90.9 kHz, 90.9 kHz)}$$

$$V_o(t) = 0.768\,\sin\left(2\pi\cdot 50000\,t - 1.01\right)\ \text{V}$$

Solution 2D.10 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 30 = 188\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 1.88 \times 10^7\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (superposition: with $V_o = 0$, $\frac{1}{sC_2}$ and $\frac{1}{sC_3}$ are in parallel from the $-$ input to ground; with $V_i = 0$, $\frac{1}{sC_1}$ and $\frac{1}{sC_3}$ are. $\frac{1}{sC_3}$ matters now, because V_- is no longer forced to 0.)

$$V_+ = 0 \quad \text{(grounded)}$$

$$V_- = \frac{\frac{1}{sC_2} \parallel \frac{1}{sC_3}}{\frac{1}{sC_1} + \frac{1}{sC_2} \parallel \frac{1}{sC_3}} V_i + \frac{\frac{1}{sC_1} \parallel \frac{1}{sC_3}}{\frac{1}{sC_2} + \frac{1}{sC_1} \parallel \frac{1}{sC_3}} V_o$$

(Only capacitors: every impedance is $\frac{1}{sC}$, so in each ratio the s cancels, e.g. $\frac{1/(sC_a)}{1/(sC_a)+1/(sC_b)} = \frac{C_b}{C_a+C_b}$, and $\frac{1}{sC_a} \parallel \frac{1}{sC_b} = \frac{1}{s(C_a+C_b)}$.) So $\alpha_+ = 0$, $\alpha_- = 0.5$, $\beta = 0.1667$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -3$, $A_0\beta = 16700$.

$$\frac{V_o}{V_i} = -3 \cdot \frac{16700}{1 + 16700} \cdot \frac{1}{1 + \dfrac{s}{3.14 \times 10^6}} = \frac{-3}{1 + \dfrac{s}{3.14 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 3.14 \times 10^6$ rad/s, $f = 500kHz$; compare $\beta\omega_t = 3.14 \times 10^6$ rad/s.) So:

$$H(s) = \frac{-9.42 \times 10^6}{s + 3.14 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 3.14 \times 10^6$; $s + 3.14 \times 10^6 = 0 \Rightarrow s = -3.14 \times 10^6$.

Factored form (leading coefficient in front):

$$H(s) = -9.42 \times 10^6 \, \frac{1}{(s + 3.14 \times 10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -3 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 500\text{k}}}, \qquad K = -3$$

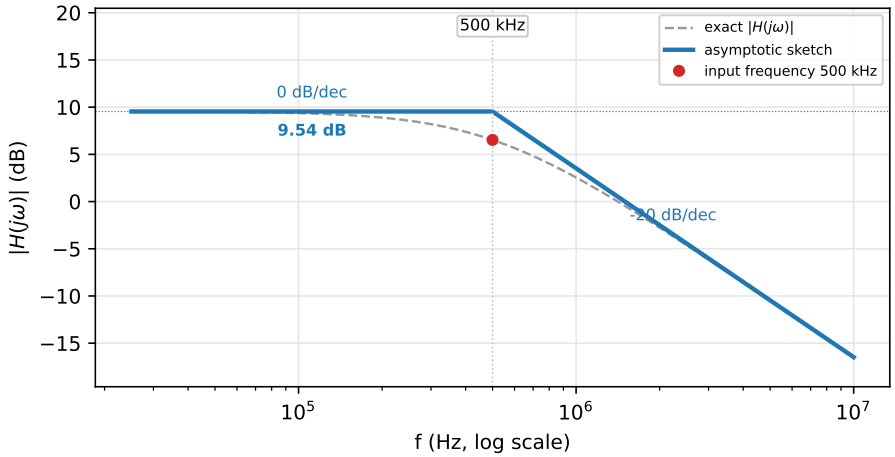
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	3.14×10^6	500 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 3 = 9.54\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 500$ kHz	0 dB/dec (flat)	9.54 dB at end
$f > 500$ kHz	-20 dB/dec	9.54 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 3 = 9.54\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin\left(2\pi \cdot 5\times10^5\, t\right)$, so $\omega_{in} = 2\pi\cdot 5\times10^5 = 3.14\times10^6$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-3	3	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 500\text{k}}}$	0.7071	-0.785
total	**2.121** (product)	$\mathbf{-3.93}$ (sum)

$$|H(j\omega_{in})| = 2.12, \qquad \angle H(j\omega_{in}) = -3.93 \text{ rad}$$

$$v_{o,ac}(t) = 2.12\times0.05\sin\left(2\pi\cdot5\times10^5\,t - 3.93\right) = 0.106\sin\left(2\pi\cdot5\times10^5\,t - 3.93\right)$$

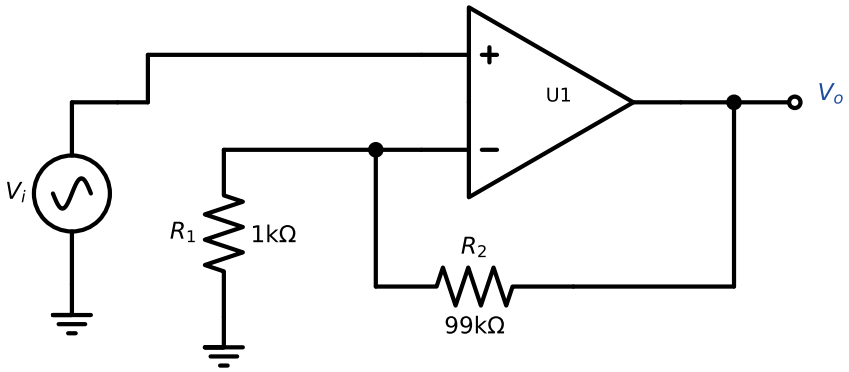
$$\boxed{V_o(t) = 0.106\sin\left(2\pi\cdot5\times10^5\,t - 3.93\right)\ \text{V}}$$

Answer

$$H(s) = -3\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500\text{k}}} \quad \text{(corners: 500 kHz)}$$

$$V_o(t) = 0.106\sin\left(2\pi\cdot5\times10^5\,t - 3.93\right)\ \text{V}$$

Solution 2D.11 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = \frac{R_1}{R_1 + R_2} V_o$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 0.01$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 100$, $A_0\beta = 1000$.

$$\frac{V_o}{V_i} = 100 \cdot \frac{1000}{1 + 1000} \cdot \frac{1}{1 + \dfrac{s}{62900}} = \frac{99.9}{1 + \dfrac{s}{62900}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 62900\,\text{rad/s}$, $f = 10kHz$; compare $\beta\omega_t = 62800\,\text{rad/s}$.) So:

$$H(s) = \frac{6.28 \times 10^6}{s + 62900}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 62900$; $s + 62900 = 0 \Rightarrow s = -62900$.

Factored form (leading coefficient in front):

$$H(s) = 6.28 \times 10^6 \, \frac{1}{(s + 62900)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 99.9 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 10\text{k}}}, \qquad K = 99.9$$

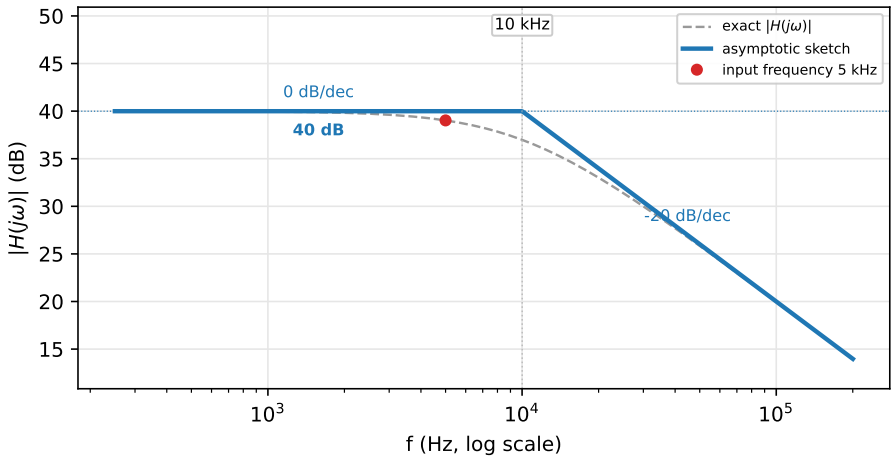
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	62900	10 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 99.9 = 40\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 10$ kHz	0 dB/dec (flat)	40 dB at end
$f > 10$ kHz	−20 dB/dec	40 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 99.9 = 40\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi \cdot 5000\,t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
99.9	99.9	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}}$	0.8946	-0.463
total	**89.37** (product)	$\mathbf{-0.463}$ (sum)

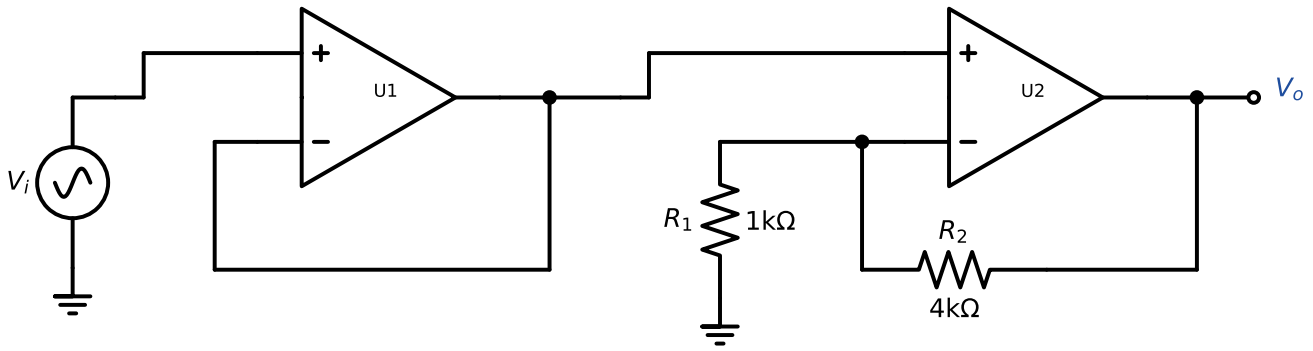
$$|H(j\omega_{in})| = 89.4, \qquad \angle H(j\omega_{in}) = -0.463 \text{ rad}$$
$$v_{o,ac}(t) = 89.4 \times 0.01 \sin(2\pi \cdot 5000\,t - 0.463) = 0.894 \sin(2\pi \cdot 5000\,t - 0.463)$$

$$\boxed{V_o(t) = 0.894 \sin(2\pi \cdot 5000\,t - 0.463) \text{ V}}$$

Answer

$$H(s) = 99.9\,\frac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}} \quad \text{(corners: 10 kHz)}$$
$$V_o(t) = 0.894 \sin(2\pi \cdot 5000\,t - 0.463) \text{ V}$$

Solution 2D.12 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 2 \times 10^5$, $\omega_b = 2\pi \cdot 5 = 31.4\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = V_{o1}$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 1$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 1$, $A_0\beta = 2 \times 10^5$.

$$\frac{V_{o1}}{V_i} = 1 \cdot \frac{2 \times 10^5}{1 + 2 \times 10^5} \cdot \frac{1}{1 + \dfrac{s}{6.28 \times 10^6}} = \frac{1}{1 + \dfrac{s}{6.28 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 6.28 \times 10^6\,\text{rad/s}$, $f = 1MHz$; compare $\beta\omega_t = 6.28 \times 10^6\,\text{rad/s}$.)
Block 2 (U2). (no current enters either input, so both are plain dividers)

$$V_+ = V_{o1}$$

$$V_- = \frac{R_1}{R_1 + R_2} V_o$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 0.2$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 5$, $A_0\beta = 40000$.

$$\frac{V_o}{V_{o1}} = 5 \cdot \frac{40000}{1 + 40000} \cdot \frac{1}{1 + \dfrac{s}{1.26 \times 10^6}} = \frac{5}{1 + \dfrac{s}{1.26 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 1.26 \times 10^6$ rad/s, $f = 200kHz$; compare $\beta\omega_t = 1.26 \times 10^6$ rad/s.) Each op-amp output is still an ideal source (it is $A(V_+ - V_-)$), so the stages multiply. Multiply the blocks and simplify:

$$H(s) = \frac{3.95 \times 10^{13}}{s^2 + 7.54 \times 10^6\, s + 7.9 \times 10^{12}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 7.54 \times 10^6\, s + 7.9 \times 10^{12}$; quadratic formula:

$$s = \frac{-7.54 \times 10^6 \pm \sqrt{(7.54 \times 10^6)^2 - 4(7.9 \times 10^{12})}}{2} = \frac{-7.54 \times 10^6 \pm 5.03 \times 10^6}{2} \Rightarrow s = -1.26{\times}10^6,\ -6.28{\times}10^6$$

Factored form (leading coefficient in front):

$$H(s) = 3.95 \times 10^{13}\, \frac{1}{(s + 1.26 \times 10^6)(s + 6.28 \times 10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 5\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 200\text{k}}}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1\text{M}}}, \qquad K = 5$$

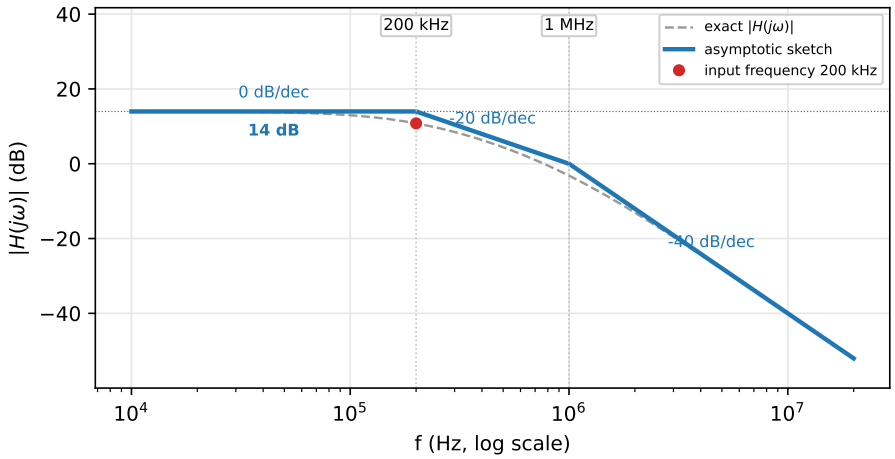
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.26×10^6	200 kHz	-20 dB/dec
pole	6.28×10^6	1 MHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 5 = 14\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 200$ kHz	0 dB/dec (flat)	14 dB at end
200 kHz – 1 MHz	-20 dB/dec	14 dB at start, -0.0000869 dB at end
$f > 1$ MHz	-40 dB/dec	-0.0000869 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin\left(2\pi\cdot 2\times 10^5\,t\right)$, so $\omega_{in} = 2\pi\cdot 2\times 10^5 = 1.26\times 10^6\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5	5	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}$	0.7071	-0.785
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{M}}}$	0.9806	-0.197
total	**3.467** (product)	**−0.983** (sum)

$$|H(j\omega_{in})| = 3.47, \qquad \angle H(j\omega_{in}) = -0.983\ \text{rad}$$
$$v_{o,ac}(t) = 3.47\times 0.1\,\sin\left(2\pi\cdot 2\times 10^5\,t - 0.983\right) = 0.347\,\sin\left(2\pi\cdot 2\times 10^5\,t - 0.983\right)$$

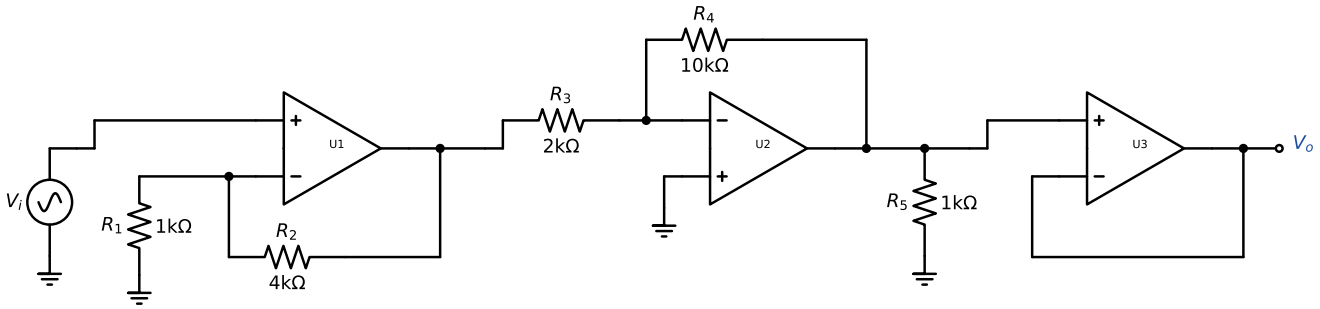
$$\boxed{V_o(t) = 0.347\,\sin\left(2\pi\cdot 2\times 10^5\,t - 0.983\right)\ \text{V}}$$

Answer

$$H(s) = 5\,\frac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{M}}} \quad \text{(corners: 200 kHz, 1 MHz)}$$

$$V_o(t) = 0.347 \sin\left(2\pi \cdot 2 \times 10^5\, t - 0.983\right) \text{ V}$$

Solution 2D.13 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

The load R_5 sits on an op-amp output, modelled as the ideal source $A(V_+ - V_-)$: it does not change the output voltage and drops out. **Block 1 (U1).** (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = \frac{R_1}{R_1 + R_2} V_{o1}$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 0.2$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = 5$, $A_0\beta = 20000$.

$$\frac{V_{o1}}{V_i} = 5 \cdot \frac{20000}{1 + 20000} \cdot \frac{1}{1 + \dfrac{s}{1.26 \times 10^6}} = \frac{5}{1 + \dfrac{s}{1.26 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 1.26\times10^6$ rad/s, $f = 200kHz$; compare $\beta\omega_t = 1.26\times10^6$ rad/s.)
Block 2 (U2). (superposition: V_{o1} alone with $V_{o2}=0$ gives a divider R_3–R_4, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_4}{R_3+R_4}V_{o1} + \frac{R_3}{R_3+R_4}V_{o2}$$

So $\alpha_+ = 0$, $\alpha_- = 0.8333$, $\beta = 0.1667$: ideal gain $G = \frac{\alpha_+-\alpha_-}{\beta} = -5$, $A_0\beta = 16700$.

$$\frac{V_{o2}}{V_{o1}} = -5\cdot\frac{16700}{1+16700}\cdot\frac{1}{1+\dfrac{s}{1.05\times10^6}} = \frac{-5}{1+\dfrac{s}{1.05\times10^6}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 1.05\times10^6$ rad/s, $f = 167kHz$; compare $\beta\omega_t = 1.05\times10^6$ rad/s.)
Block 3 (U3). (no current enters either input, so both are plain dividers)

$$V_+ = V_{o2}$$
$$V_- = V_o$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 1$: ideal gain $G = \frac{\alpha_+-\alpha_-}{\beta} = 1$, $A_0\beta = 10^5$.

$$\frac{V_o}{V_{o2}} = 1\cdot\frac{10^5}{1+10^5}\cdot\frac{1}{1+\dfrac{s}{6.28\times10^6}} = \frac{1}{1+\dfrac{s}{6.28\times10^6}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 6.28\times10^6$ rad/s, $f = 1MHz$; compare $\beta\omega_t = 6.28\times10^6$ rad/s.)
Each op-amp output is still an ideal source (it is $A(V_+ - V_-)$), so the stages multiply. Multiply the blocks and simplify:

$$H(s) = \frac{-2.07\times10^{20}}{s^3 + 8.59\times10^6\, s^2 + 1.58\times10^{13}\, s + 8.27\times10^{18}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s=-p$ of the denominator is a *pole*, every root $s=-z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^3+8.59\times10^6\, s^2+1.58\times10^{13}\, s+8.27\times10^{18}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -1.05\times10^6,\ -1.26\times10^6,\ -6.28\times10^6$.

Factored form (leading coefficient in front):

$$H(s) = -2.07\times10^{20}\,\frac{1}{(s+1.05\times10^6)(s+1.26\times10^6)(s+6.28\times10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -25\,\frac{1}{1+\dfrac{s}{2\pi\cdot167\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot200\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot1\text{M}}}, \qquad K=-25$$

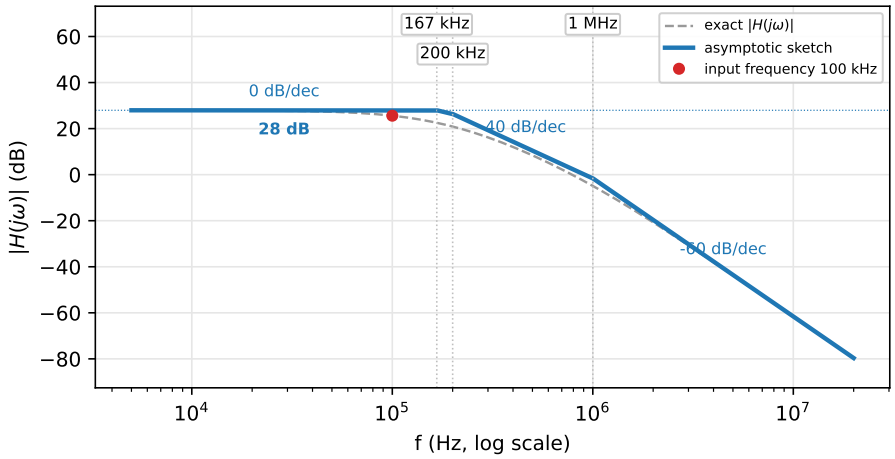
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.05×10^6	167 kHz	−20 dB/dec
pole	1.26×10^6	200 kHz	−20 dB/dec
pole	6.28×10^6	1 MHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 25 = 28\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 167$ kHz	0 dB/dec (flat)	28 dB at end
167 kHz – 200 kHz	-20 dB/dec	28 dB at start, 26.4 dB at end
200 kHz – 1 MHz	-40 dB/dec	26.4 dB at start, -1.58 dB at end
$f > 1$ MHz	-60 dB/dec	-1.58 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 25 = 28\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.02\sin\left(2\pi \cdot 10^5\, t\right)$, so $\omega_{in} = 2\pi \cdot 10^5 = 6.28 \times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-25	25	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 167\text{k}}}$	0.8575	-0.54
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}$	0.8944	-0.464
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{M}}}$	0.995	-0.0997
total	**19.08** (product)	$-$**4.25** (sum)

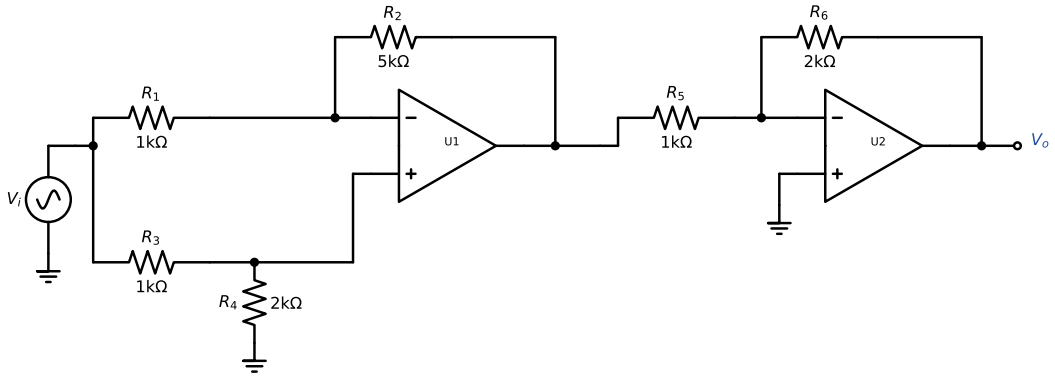
$$|H(j\omega_{in})| = 19.1, \qquad \angle H(j\omega_{in}) = -4.25 \text{ rad}$$
$$v_{o,ac}(t) = 19.1 \times 0.02 \sin\left(2\pi\cdot 10^5\, t - 4.25\right) = 0.382 \sin\left(2\pi\cdot 10^5\, t - 4.25\right)$$

$$\boxed{V_o(t) = 0.382 \sin\left(2\pi\cdot 10^5\, t - 4.25\right) \text{ V}}$$

Answer

$$H(s) = -25\,\frac{1}{1+\dfrac{s}{2\pi\cdot 167\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 200\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{M}}} \quad \text{(corners: 167 kHz, 200 kHz, 1 MHz)}$$
$$V_o(t) = 0.382 \sin\left(2\pi\cdot 10^5\, t - 4.25\right) \text{ V}$$

Solution 2D.14 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 20 = 126\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 1.26 \times 10^7\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (each input is a divider; the $-$ input sees both sources, use superposition)

$$V_+ = \frac{R_4}{R_3 + R_4} V_i$$
$$V_- = \frac{R_2}{R_1 + R_2} V_i + \frac{R_1}{R_1 + R_2} V_{o1}$$

So $\alpha_+ = 0.6667$, $\alpha_- = 0.8333$, $\beta = 0.1667$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -1$, $A_0\beta = 16700$.

$$\frac{V_{o1}}{V_i} = -1 \cdot \frac{16700}{1 + 16700} \cdot \frac{1}{1 + \dfrac{s}{2.09 \times 10^6}} = \frac{-0.9999}{1 + \dfrac{s}{2.09 \times 10^6}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 2.09\times10^6$ rad/s, $f = 333kHz$; compare $\beta\omega_t = 2.09\times10^6$ rad/s.)
Block 2 (U2). (superposition: V_{o1} alone with $V_o = 0$ gives a divider R_5–R_6, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_6}{R_5+R_6}V_{o1} + \frac{R_5}{R_5+R_6}V_o$$

So $\alpha_+ = 0$, $\alpha_- = 0.6667$, $\beta = 0.3333$: ideal gain $G = \frac{\alpha_+-\alpha_-}{\beta} = -2$, $A_0\beta = 33300$.

$$\frac{V_o}{V_{o1}} = -2\cdot\frac{33300}{1+33300}\cdot\frac{1}{1+\dfrac{s}{4.19\times10^6}} = \frac{-2}{1+\dfrac{s}{4.19\times10^6}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 4.19\times10^6$ rad/s, $f = 667kHz$; compare $\beta\omega_t = 4.19\times10^6$ rad/s.)
Each op-amp output is still an ideal source (it is $A(V_+ - V_-)$), so the stages multiply. Multiply the blocks and simplify:

$$H(s) = \frac{1.75\times10^{13}}{s^2 + 6.28\times10^6\, s + 8.77\times10^{12}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 6.28\times10^6\, s + 8.77\times10^{12}$; quadratic formula:

$$s = \frac{-6.28\times10^6 \pm \sqrt{(6.28\times10^6)^2 - 4(8.77\times10^{12})}}{2} = \frac{-6.28\times10^6 \pm 2.09\times10^6}{2} \Rightarrow s = -2.09{\times}10^6,\ -4.19{\times}10^6$$

Factored form (leading coefficient in front):

$$H(s) = 1.75\times10^{13}\,\frac{1}{(s+2.09\times10^6)(s+4.19\times10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 2\,\frac{1}{1+\dfrac{s}{2\pi\cdot 333\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 667\text{k}}}, \qquad K = 2$$

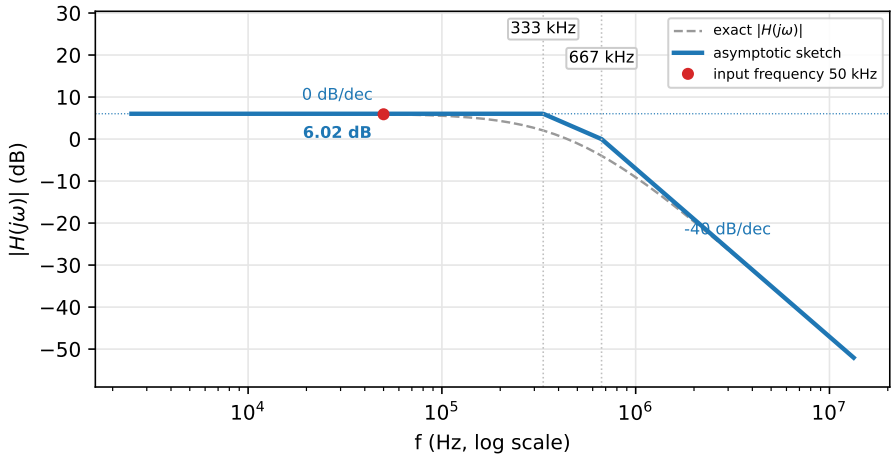
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2.09×10^6	333 kHz	−20 dB/dec
pole	4.19×10^6	667 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 2 = 6.02$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 333$ kHz	0 dB/dec (flat)	6.02 dB at end
333 kHz – 667 kHz	−20 dB/dec	6.02 dB at start, -0.000521 dB at end
$f > 667$ kHz	−40 dB/dec	-0.000521 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 2 = 6.02$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin\left(2\pi\cdot 50000\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi\cdot 50000 = 3.14\times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
2	2	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 333\text{k}}}$	0.9889	−0.149
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 667\text{k}}}$	0.9972	−0.0749
total	**1.972** (product)	**−0.224** (sum)

$$|H(j\omega_{in})| = 1.97, \qquad \angle H(j\omega_{in}) = -0.224 \text{ rad}$$
$$v_{o,ac}(t) = 1.97\times 0.05\,\sin\left(2\pi\cdot 50000\,t + 0.562\right) = 0.0986\,\sin\left(2\pi\cdot 50000\,t + 0.562\right)$$

(phase: $\theta + \angle H = \frac{\pi}{4} + (-0.224) = 0.562$ rad.)

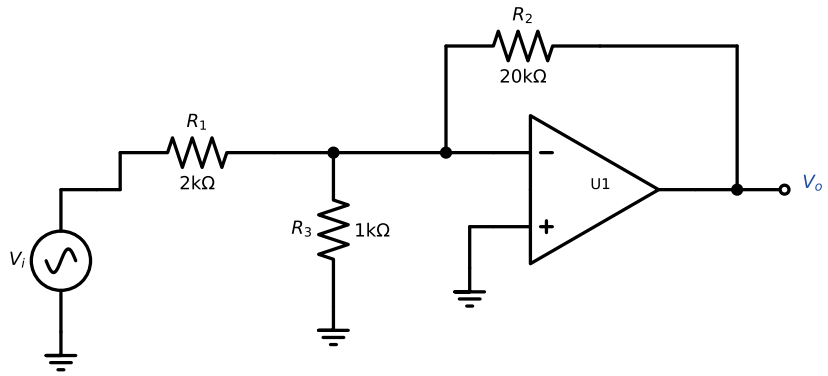
$$\boxed{V_o(t) = 0.0986\,\sin\left(2\pi\cdot 50000\,t + 0.562\right)\ \text{V}}$$

Answer

$$H(s) = 2\,\frac{1}{1+\dfrac{s}{2\pi\cdot 333\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 667\text{k}}} \quad \text{(corners: 333 kHz, 667 kHz)}$$

$$V_o(t) = 0.0986\,\sin\left(2\pi\cdot 50000\,t + 0.562\right)\ \text{V}$$

Solution 2D.15 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (superposition: with $V_o = 0$, R_2 and R_3 are in parallel from the $-$ input to ground; with $V_i = 0$, R_1 and R_3 are. R_3 matters now, because V_- is no longer forced to 0.)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_2 \parallel R_3}{R_1 + R_2 \parallel R_3} V_i + \frac{R_1 \parallel R_3}{R_2 + R_1 \parallel R_3} V_o$$

So $\alpha_+ = 0$, $\alpha_- = 0.3226$, $\beta = 0.03226$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -10$, $A_0\beta = 3230$.

$$\frac{V_o}{V_i} = -10 \cdot \frac{3230}{1 + 3230} \cdot \frac{1}{1 + \dfrac{s}{2.03 \times 10^5}} = \frac{-9.997}{1 + \dfrac{s}{2.03 \times 10^5}}$$

(closed-loop bandwidth $\omega_b(1 + A_0\beta) = 2.03 \times 10^5$ rad/s, $f = 32.3kHz$; compare $\beta\omega_t = 2.03 \times 10^5$ rad/s.) So:

$$H(s) = \frac{-2.03 \times 10^6}{s + 2.03 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 2.03 \times 10^5$; $s + 2.03 \times 10^5 = 0 \Rightarrow s = -2.03 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = -2.03 \times 10^6 \, \frac{1}{(s + 2.03 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -10 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 32.3\text{k}}}, \qquad K = -10$$

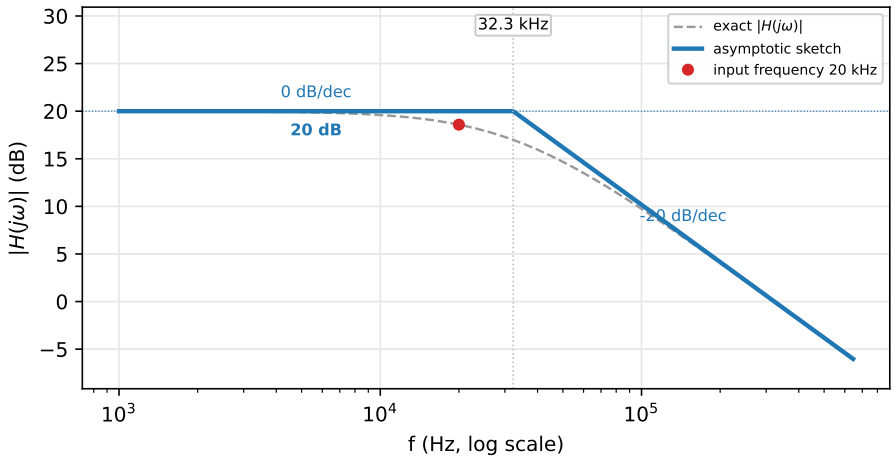
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2.03×10^5	32.3 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 32.3$ kHz	0 dB/dec (flat)	20 dB at end
$f > 32.3$ kHz	-20 dB/dec	20 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 10 = 20\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\cos(2\pi \cdot 20000\,t)$, so $\omega_{in} = 2\pi \cdot 20000 = 1.26 \times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-10	9.997	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 32.3\text{k}}}$	0.85	-0.555
total	**8.497** (product)	$-$**3.7** (sum)

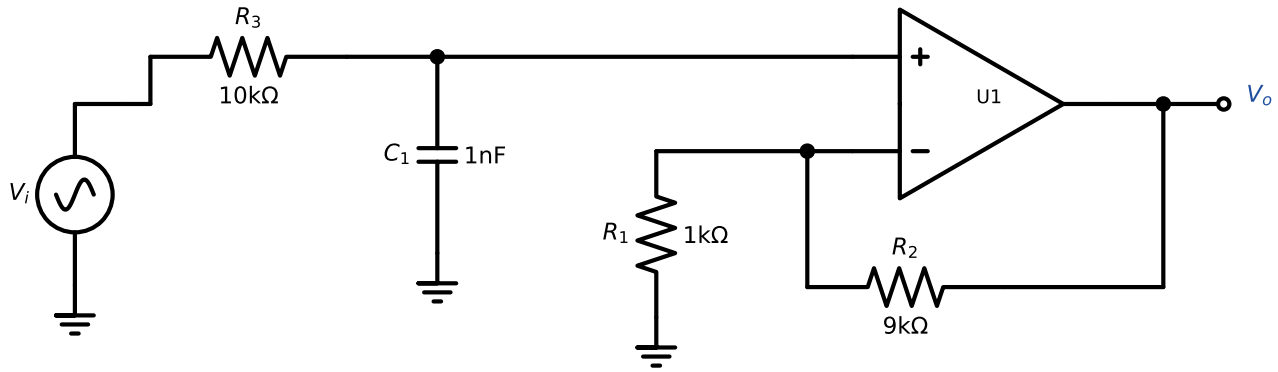
$$|H(j\omega_{in})| = 8.5, \qquad \angle H(j\omega_{in}) = -3.7 \text{ rad}$$
$$v_{o,ac}(t) = 8.5 \times 0.05\,\cos(2\pi \cdot 20000\,t - 3.7) = 0.425\,\cos(2\pi \cdot 20000\,t - 3.7)$$

$$\boxed{V_o(t) = 0.425\,\cos(2\pi \cdot 20000\,t - 3.7)\ \text{V}}$$

Answer

$$H(s) = -10\,\frac{1}{1+\dfrac{s}{2\pi \cdot 32.3\text{k}}} \quad \text{(corners: 32.3 kHz)}$$
$$V_o(t) = 0.425\,\cos(2\pi \cdot 20000\,t - 3.7)\ \text{V}$$

Solution 2D.16 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = \frac{\frac{1}{sC_1}}{R_3 + \frac{1}{sC_1}} V_i$$

$$V_- = \frac{R_1}{R_1 + R_2} V_o$$

Here α and β depend on s, so substitute everything and simplify:

$$\frac{V_o}{V_i} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{6.28 \times 10^{11}}{s^2 + 7.28 \times 10^5\, s + 6.28 \times 10^{10}}$$

So:

$$H(s) = \frac{6.28 \times 10^{11}}{s^2 + 7.28 \times 10^5\, s + 6.28 \times 10^{10}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 7.28 \times 10^5\, s + 6.28 \times 10^{10}$; quadratic formula:

$$s = \frac{-7.28 \times 10^5 \pm \sqrt{(7.28 \times 10^5)^2 - 4(6.28 \times 10^{10})}}{2} = \frac{-7.28 \times 10^5 \pm 5.28 \times 10^5}{2} \Rightarrow s = -10^5,\ -6.28{\times}10^5$$

Factored form (leading coefficient in front):

$$H(s) = 6.28 \times 10^{11}\, \frac{1}{(s + 10^5)(s + 6.28 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 10\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 15.9\text{k}}}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 100\text{k}}}, \qquad K = 10$$

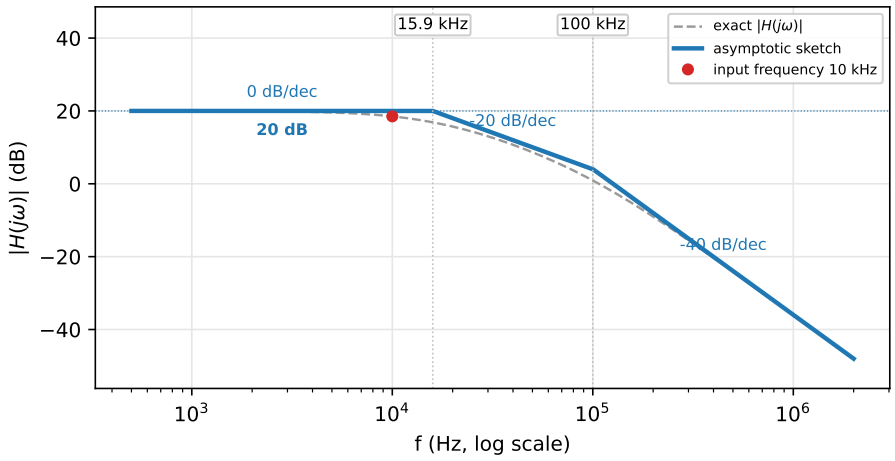
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10^5	15.9 kHz	−20 dB/dec
pole	6.28×10^5	100 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 15.9 kHz	0 dB/dec (flat)	20 dB at end
15.9 kHz – 100 kHz	−20 dB/dec	20 dB at start, 4.03 dB at end
$f >$ 100 kHz	−40 dB/dec	4.03 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 10 = 20\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 10000\,t)$, so $\omega_{in} = 2\pi \cdot 10000 = 62800$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	9.999	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 15.9\text{k}}}$	0.8467	−0.561
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100\text{k}}}$	0.995	−0.0997
total	**8.424** (product)	**−0.661** (sum)

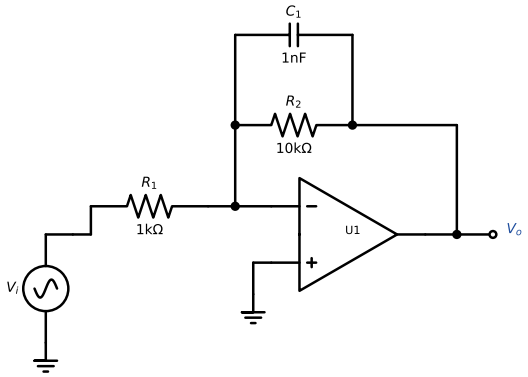
$$|H(j\omega_{in})| = 8.42, \qquad \angle H(j\omega_{in}) = -0.661 \text{ rad}$$
$$v_{o,ac}(t) = 8.42 \times 0.1 \sin(2\pi \cdot 10000\,t - 0.661) = 0.842 \sin(2\pi \cdot 10000\,t - 0.661)$$

$$\boxed{V_o(t) = 0.842 \sin(2\pi \cdot 10000\,t - 0.661) \text{ V}}$$

Answer

$$H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi \cdot 15.9\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 100\text{k}}} \quad \text{(corners: 15.9 kHz, 100 kHz)}$$
$$V_o(t) = 0.842 \sin(2\pi \cdot 10000\,t - 0.661) \text{ V}$$

Solution 2D.17 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (superposition: V_i alone with $V_o = 0$ gives a divider R_1–Z_{f1}, and vice versa)

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{10^9}{s + 10^5}$$

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{Z_{f1}}{R_1 + Z_{f1}} V_i + \frac{R_1}{R_1 + Z_{f1}} V_o$$

Here α and β depend on s, so substitute everything and simplify:

$$\frac{V_o}{V_i} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{-6.28 \times 10^{12}}{s^2 + 7.38 \times 10^6\, s + 6.28 \times 10^{11}}$$

So:

$$H(s) = \frac{-6.28 \times 10^{12}}{s^2 + 7.38 \times 10^6\, s + 6.28 \times 10^{11}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 7.38 \times 10^6\, s + 6.28 \times 10^{11}$; quadratic formula:

$$s = \frac{-7.38 \times 10^6 \pm \sqrt{(7.38 \times 10^6)^2 - 4(6.28 \times 10^{11})}}{2} = \frac{-7.38 \times 10^6 \pm 7.21 \times 10^6}{2} \Rightarrow s = -86100,\ -7.3{\times}10^6$$

Factored form (leading coefficient in front):

$$H(s) = -6.28 \times 10^{12}\, \frac{1}{(s + 86100)(s + 7.3 \times 10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -10\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 13.7\text{k}}}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.16\text{M}}}, \qquad K = -10$$

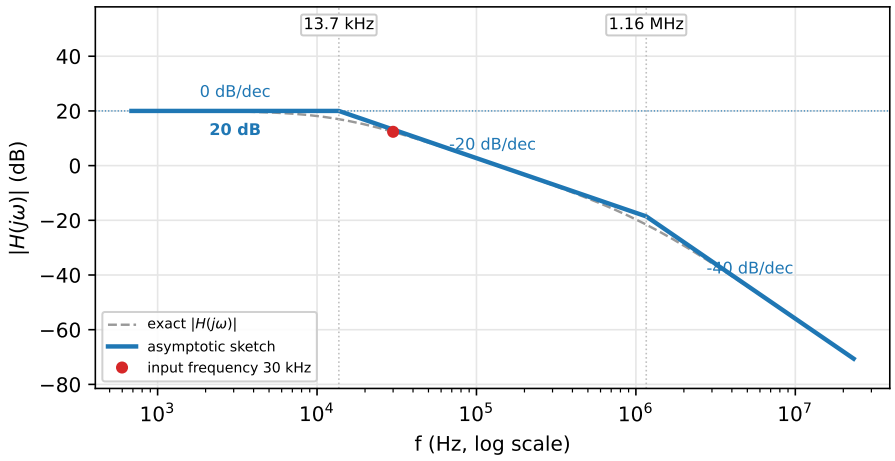
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	86100	13.7 kHz	−20 dB/dec
pole	7.3×10^6	1.16 MHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 13.7 kHz	0 dB/dec (flat)	20 dB at end
13.7 kHz – 1.16 MHz	−20 dB/dec	20 dB at start, -18.6 dB at end
$f >$ 1.16 MHz	−40 dB/dec	-18.6 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 10 = 20\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin\left(2\pi\cdot 30000\,t - \frac{\pi}{6}\right)$, so $\omega_{in} = 2\pi\cdot 30000 = 1.88\times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-10	9.999	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 13.7\text{k}}}$	0.4155	-1.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.16\text{M}}}$	0.9997	-0.0258
total	**4.154** (product)	$-$**4.31** (sum)

$$|H(j\omega_{in})| = 4.15, \qquad \angle H(j\omega_{in}) = -4.31 \text{ rad}$$
$$v_{o,ac}(t) = 4.15\times 0.1\,\sin\left(2\pi\cdot 30000\,t - 4.83\right) = 0.415\,\sin\left(2\pi\cdot 30000\,t - 4.83\right)$$

(phase: $\theta + \angle H = -\frac{\pi}{6} + (-4.31) = -4.83$ rad.)

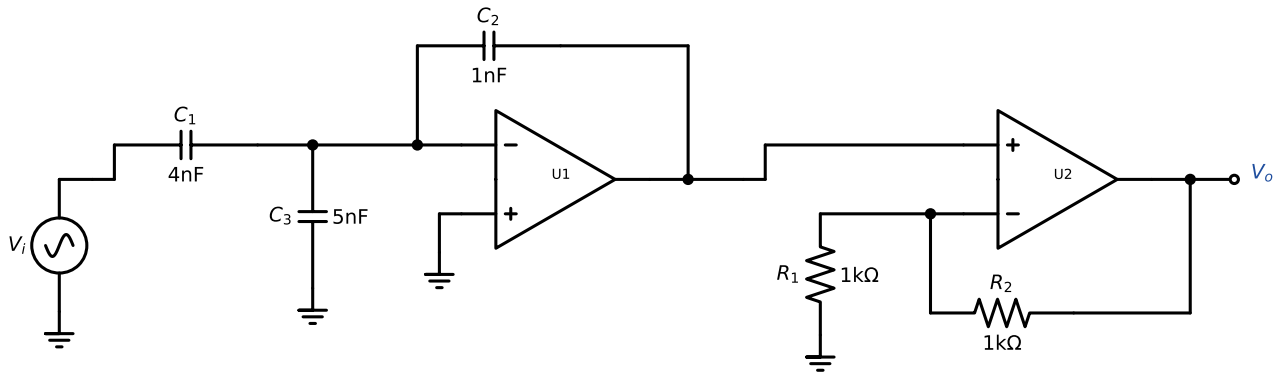
$$\boxed{V_o(t) = 0.415\,\sin\left(2\pi\cdot 30000\,t - 4.83\right)\ \text{V}}$$

Answer

$$H(s) = -10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 13.7\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.16\text{M}}} \qquad \text{(corners: 13.7 kHz, 1.16 MHz)}$$

$$V_o(t) = 0.415 \sin\left(2\pi \cdot 30000\, t - 4.83\right) \text{ V}$$

Solution 2D.18 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp model

Non-ideal op-amp model (tool T5): $V_{out} = A(s)\,(V_+ - V_-)$ with $A(s) = \dfrac{A_0}{1 + s/\omega_b}$, $A_0 = 10^5$, $\omega_b = 2\pi \cdot 10 = 62.8\,\text{rad/s}$, so the unity-gain frequency is $\omega_t = A_0\omega_b = 6.28 \times 10^6\,\text{rad/s}$. The virtual short is *not* used; only "no current into the inputs" is still true.

Step 4 — Transfer function, op-amp by op-amp

Recipe for each op-amp: write V_+ and V_- by superposition of the block input and the op-amp output (each alone is a plain divider): $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$. Then $V_{out} = A(\alpha_+ - \alpha_-)V_{in} - A\beta V_{out}$, i.e.

$$\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{G}{1 + \dfrac{1}{A\beta}}, \qquad G = \frac{\alpha_+ - \alpha_-}{\beta} \text{ (the ideal gain)}$$

With $A(s)$ inserted and constant α, β:

$$\frac{V_{out}}{V_{in}} = G\,\frac{A_0\beta}{1 + A_0\beta} \cdot \frac{1}{1 + \dfrac{s}{\omega_b(1 + A_0\beta)}}, \qquad \omega_b(1 + A_0\beta) \approx \beta\,\omega_t$$

Block 1 (U1). (superposition: with $V_{o1} = 0$, $\frac{1}{sC_2}$ and $\frac{1}{sC_3}$ are in parallel from the $-$ input to ground; with $V_i = 0$, $\frac{1}{sC_1}$ and $\frac{1}{sC_3}$ are. $\frac{1}{sC_3}$ matters now, because V_- is no longer forced to 0.)

$$V_+ = 0 \quad \text{(grounded)}$$

$$V_- = \frac{\frac{1}{sC_2} \parallel \frac{1}{sC_3}}{\frac{1}{sC_1} + \frac{1}{sC_2} \parallel \frac{1}{sC_3}} V_i + \frac{\frac{1}{sC_1} \parallel \frac{1}{sC_3}}{\frac{1}{sC_2} + \frac{1}{sC_1} \parallel \frac{1}{sC_3}} V_{o1}$$

(Only capacitors: every impedance is $\frac{1}{sC}$, so in each ratio the s cancels, e.g. $\frac{1/(sC_a)}{1/(sC_a)+1/(sC_b)} = \frac{C_b}{C_a+C_b}$, and $\frac{1}{sC_a} \parallel \frac{1}{sC_b} = \frac{1}{s(C_a+C_b)}$.) So $\alpha_+ = 0$, $\alpha_- = 0.4$, $\beta = 0.1$: ideal gain $G = \frac{\alpha_+ - \alpha_-}{\beta} = -4$, $A_0\beta = 10000$.

$$\frac{V_{o1}}{V_i} = -4 \cdot \frac{10000}{1 + 10000} \cdot \frac{1}{1 + \dfrac{s}{6.28 \times 10^5}} = \frac{-4}{1 + \dfrac{s}{6.28 \times 10^5}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 6.28\times10^5$ rad/s, $f = 100kHz$; compare $\beta\omega_t = 6.28\times10^5$ rad/s.)
Block 2 (U2). (no current enters either input, so both are plain dividers)

$$V_+ = V_{o1}$$
$$V_- = \frac{R_1}{R_1+R_2}V_o$$

So $\alpha_+ = 1$, $\alpha_- = 0$, $\beta = 0.5$: ideal gain $G = \frac{\alpha_+-\alpha_-}{\beta} = 2$, $A_0\beta = 50000$.

$$\frac{V_o}{V_{o1}} = 2\cdot\frac{50000}{1+50000}\cdot\frac{1}{1+\dfrac{s}{3.14\times10^6}} = \frac{2}{1+\dfrac{s}{3.14\times10^6}}$$

(closed-loop bandwidth $\omega_b(1+A_0\beta) = 3.14\times10^6$ rad/s, $f = 500kHz$; compare $\beta\omega_t = 3.14\times10^6$ rad/s.)
Each op-amp output is still an ideal source (it is $A(V_+ - V_-)$), so the stages multiply. Multiply the blocks and simplify:

$$H(s) = \frac{-1.58\times10^{13}}{s^2 + 3.77\times10^6\, s + 1.97\times10^{12}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s=-p$ of the denominator is a *pole*, every root $s=-z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 3.77\times10^6\, s + 1.97\times10^{12}$; quadratic formula:

$$s = \frac{-3.77\times10^6 \pm \sqrt{(3.77\times10^6)^2 - 4(1.97\times10^{12})}}{2} = \frac{-3.77\times10^6 \pm 2.51\times10^6}{2} \Rightarrow s = -6.28{\times}10^5,\ -3.14{\times}10^6$$

Factored form (leading coefficient in front):

$$H(s) = -1.58\times10^{13}\,\frac{1}{(s+6.28\times10^5)(s+3.14\times10^6)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -8\,\frac{1}{1+\dfrac{s}{2\pi\cdot100\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot500\text{k}}}, \qquad K = -8$$

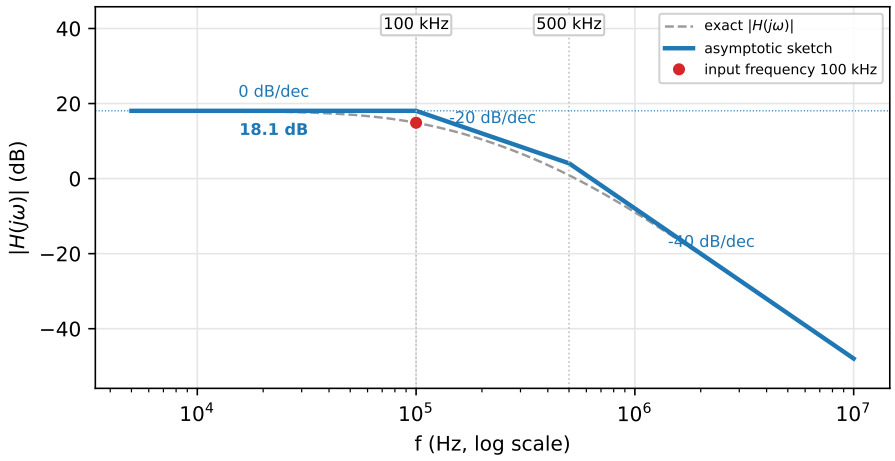
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	6.28×10^5	100 kHz	−20 dB/dec
pole	3.14×10^6	500 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 8 = 18.1$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 100$ kHz	0 dB/dec (flat)	18.1 dB at end
100 kHz – 500 kHz	-20 dB/dec	18.1 dB at start, 4.08 dB at end
$f > 500$ kHz	-40 dB/dec	4.08 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 8 = 18.1$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.02\sin\left(2\pi\cdot 10^5\, t\right)$, so $\omega_{in} = 2\pi\cdot 10^5 = 6.28\times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-8	7.999	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$	0.7071	-0.785
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 500\text{k}}}$	0.9806	-0.197
total	**5.547** (product)	$\mathbf{-4.12}$ (sum)

$$|H(j\omega_{in})| = 5.55, \qquad \angle H(j\omega_{in}) = -4.12 \text{ rad}$$

$$v_{o,ac}(t) = 5.55\times 0.02\,\sin\left(2\pi\cdot 10^5\, t - 4.12\right) = 0.111\,\sin\left(2\pi\cdot 10^5\, t - 4.12\right)$$

$$\boxed{V_o(t) = 0.111\,\sin\left(2\pi\cdot 10^5\, t - 4.12\right)\ \text{V}}$$

Answer

$$H(s) = -8\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500\text{k}}} \quad \text{(corners: 100 kHz, 500 kHz)}$$

$$V_o(t) = 0.111\,\sin\left(2\pi\cdot 10^5\,t - 4.12\right)\ \text{V}$$